# Self-Justification Ex Machina

jpt4*

In memory of Dan E. Willard (1948–2023)

**Abstract**

Sebastian Oberhoff's essay *Incompleteness Ex Machina* proves Gödel's incompleteness theorems twice, once with sentences and once with programs, and concludes that Gödel's work divides into two parts: teaching arithmetic to compute, and the incompleteness theorems, which are victory laps. We look closely at the one step of the second theorem that Oberhoff takes "with some faith": that a system which can watch any run of its own proof-searcher can also prove, of the run that never happens, that it would have seen it. Willard's self-justifying axiom systems are what remains when that step is withheld. His 2005 system $\mathrm{IS}_D(A)$, for semantic tableaux and for tableau lists with one-quantifier-block lemmas, proves every true $\Sigma_1$-class sentence of its language and takes its own Level(1) consistency as an axiom. Willard proved, in ordinary metamathematics, that it is consistent whenever the $\Pi_1$-class theorems of its base theory are true. His 2001 system $\mathrm{IS}(A)$ asserts the weaker Level(0−) consistency and is consistent under the analogous hypothesis of regular consistency. Under its hypothesis, and with one reading of Willard's pairing of a sentence with its negation, $\mathrm{IS}_D(A)$ is honest; if its base theory also proves the $\Pi_1$-class theorems of Peano Arithmetic (with Willard's function symbols), it obeys the first incompleteness theorem. Willard reported in 1994 that $\mathrm{IS}(A)$ satisfies the first incompleteness theorem but not the second. We state everything at the granularity self-justification needs (language, deduction method, consistency notion, proof predicate, encoding), and retell Willard's 2005 consistency proof as a program in Oberhoff's case format. We then show that in a system that asserts its own Level(1) consistency, under stated conditions on its proof predicate, fixed point and encoding, the faith step, aimed at the sentence the consistency axiom pairs with the Gödel sentence, is provably equivalent to the Gödel sentence, hence unprovable. For $\mathrm{IS}_D(A)$ this rests on our reading of Willard's encoding, on his assertion that his proof predicate is correct, and on one uniqueness property of substitution that we transpose from his 2001 paper. Willard's 2005 Theorem 6 then shows the faith being bought: a bounded transcript form of the step is a $\Pi_1$-class theorem of Peano Arithmetic with Willard's function symbols, and one axiom asserting that a single large number exists makes the system inconsistent. Its fourth step, as printed, identifies two different sentences, and we note what a repair needs. Gödel's work, viewed this way, splits into three parts, not two.

> Another way to motivate the axiom system IS is to consider a philosopher's response to the question of whether multiplication is a function. His first answer would likely be "of course yes". But if one then asked the philosopher to construct a sequence of natural numbers $a_0, a_1, a_2, a_3, \ldots$ where $a_0 \geq 2$ and $a_{i+1} = (a_i)^2$, a hedge would quickly follow because $a_n$ requires at least $2^n$ bits. The expected response would then be that "multiplication is a function in theory but not always in practice". — Dan Willard, SUNY Albany TR 93-10, p. 6

*Revision of 25 October 2026, prepared with Claude (Anthropic) in the `sjas` research repository. Willard's results are cited from that repository's codification of his papers (`nachlass/codification/codified-sjas.md`), with page numbers referring to the author copies held there and proof statuses as recorded there.

# 1 Victory Laps

Oberhoff's essay [Oberhoff 2018] does something deceptively simple. It writes out Gödel's first incompleteness theorem, Rosser's improvement of it, and the second incompleteness theorem as explicit case analyses. Then it replaces Gödel's sentence $\mathsf{G}$ ("I am not provable") by $\ddot{\mathsf{G}}$ = «$\mathsf{P}(\mathsf{P})$ doesn't halt», where $\mathsf{P}$ is the program

```
P(A) {
  for s ∈ all possible strings {
    if s is a proof in 𝓕 that A(A) doesn't halt {
      halt;
    }
  }
}
```

and observes that each case analysis can be copied almost word for word. The punchline (his §5) is that $\ddot{\mathsf{G}}$ is $\mathsf{G}$ wearing Clark Kent's glasses: $\mathsf{P}(\mathsf{P})$ is searching $\mathcal{F}$'s proofs for a proof of $\ddot{\mathsf{G}}$, so $\ddot{\mathsf{G}}$ says "I am not provable". His moral (§6) is a refactoring. The hard part of Gödel's work was teaching arithmetic to compute, packaged as

> **Theorem 7** (Turing Completeness of Elementary Arithmetic). *Every formal system capable of reasoning about elementary arithmetic can also reason about algorithms,*

which Oberhoff leaves "to smarter people". The incompleteness theorems themselves are victory laps.

For the first incompleteness theorem we accept this completely. This essay is about the second. Its proof in [Oberhoff 2018] contains exactly one step that Oberhoff marks as taken on trust (Theorem 6, third step):

> Now we simply observe (with some faith) that there wasn't actually anything in the reasoning we used to prove this that we couldn't carry out just as well in $\mathcal{F}$.

For the systems Oberhoff has in mind, that faith is well placed. But some formal systems can reason about algorithms in every sense the first theorem needs, and for them the faith is misplaced: they prove their own consistency, and, by theorems of Willard's proved in ordinary metamathematics, they are consistent whenever their base theory has a bounded proof predicate and true $\Pi_1$-class theorems. For 2001 this is Willard's "regular consistency" (2001, p. 12); Theorem H gives the two cases exactly. Dan Willard constructed them in papers from 1993 to 2020 [Willard TR 93-10, Willard 2001, Willard 2005, Willard 2006a, Willard 2020]. (We cite Willard's papers by year, as in the bibliography: "2002a" is [Willard 2002a].) This essay tells their story in Oberhoff's language. Our headline is that the faith step is not part of the victory lap. It is a third part of Gödel's work, and it has a price.

**The standard of precision.** Oberhoff's essay is a paper proof at the granularity the second incompleteness theorem needs: for Peano Arithmetic and its relatives, it does not matter which language, which proof system, which notion of consistency or which Gödel numbering one picks, since the system proves that the choices agree. Self-justification lives exactly where those choices stop agreeing. Willard's systems are self-justifying for one deduction method and inconsistent for another, consistent at one consistency level and not at the next, and his proofs measure Gödel numbers in bits. An exegesis that is to have the proof status of Oberhoff's must therefore fix every one of these parameters, and say where a result depends on them. Section 2 fixes them. Throughout,

"Willard proved" means that the result is printed with a complete proof (status `full`); other statuses are named where they occur. Results labelled *ours* are proved here.

## 2 Conventions

**Configurations.** Following Willard, a *configuration* $(\alpha, D)$ is an axiom system $\alpha$ together with a deduction method $D$; $\alpha \vdash_D \varphi$ means that $\varphi$ has a $D$-proof from $\alpha$. $D$ is *$\alpha$-sound* if $\alpha \vdash_D \varphi$ implies that $\varphi$ holds in every model of $\alpha$, and *$\alpha$-complete* if the converse holds (2001, §7, p. 37). All deduction methods in this essay (Hilbert-style deduction, semantic tableaux, and Willard's tableau lists below) are sound and complete in this sense. They differ in proof *length*, and self-justification is a phenomenon of length.

**Fact A** (Willard 2001, Lemma 7.1B, p. 37, `full`)**.** *If $D$ is $\alpha$-sound and $\alpha$-complete, $\alpha \vdash_D \varphi$, $\alpha \vdash_D \psi$, and $\varphi \wedge \psi \supset \chi$ holds in every model of $\alpha$, then $\alpha \vdash_D \chi$.*

So the theorems of $(\alpha, D)$ are closed under logical consequence. What is not closed is the size of proofs: for a cut-free $D$, Willard notes, the combined proof "could then correspond to a bit string more than exponentially larger than the combined lengths of the three initial proofs" (2001, p. 38, text layer).

**Languages.** Three languages occur.

- The language of arithmetic, with 0, successor, $+$ and $\times$, and the usual classes $\Delta_0$, $\Pi_1$, $\Sigma_1$.
- Willard's 2001 language: the seven *grounding functions* (truncated subtraction, integer division, predecessor, maximum, logarithm, root, and *Count*, all *non-growth*: $F(\vec{a}) \leq \max(\vec{a})$) and one constant symbol $\bar{n}$ for every natural number $n$ (2001, pp. 9–10). Its classes are written $\Delta_0^-$, $\Pi_1^-$, $\Sigma_1^-$. Addition is not a function symbol; Group-1 of IS($A$) asserts its totality by the $\Pi_2^-$ axiom $\forall x \, \forall y \, \exists z \; z - x = y$ (2001, Eq. 6).
- Willard's 2005 *U-Grounding* language: six grounding functions plus the growth functions $+$ and $\mathit{Double}(x) = x + x$, eight function symbols in all, with constants $\bar{c}_0$, $\bar{c}_1$ for 0 and 1 (2005, §2, p. 4, and §3, p. 8). Every $n \geq 2$ is named by a term $\overline{n}$, Willard's "U-Grounded Binary Representation", with at most $2 \log_2 n$ occurrences of $+$ and *Double* applied to 1 (p. 4, where the overline is Willard's notation). Its classes are $\Delta_0^*$, $\Pi_1^*$, $\Sigma_1^*$; bounded quantifiers have the form $\forall v \leq t$ and $\exists v \leq t$ (p. 5). A negation written in front of unbounded quantifiers yields a sentence in neither class: for a proper $\Sigma_1^*$ sentence $\Upsilon$, "$\neg\Upsilon$" is "technically viewed ... as not constituting a formal $\Pi_1^*$ sentence", although it is logically equivalent to one (Remark 5, II, p. 27, text layer). Multiplication is not a function symbol; its graph $M(x, y, z)$ is a $\Delta_0^*$ formula (2005, Eq. 4), so the system knows multiplication's algebra but not its totality. Multiplication by any fixed constant is total, through iterated addition; what is withheld is the *uniform* operation $x \cdot y$ (codified statement §3.1).

We write "bounded", "$\Pi_1$-class" and "$\Sigma_1$-class" for whichever of these classes is in force. A bar over a number, $\bar{n}$, denotes the constant for $n$ in the 2001 language and the term $\overline{n}$ in the 2005 language; no argument below needs a constant for every number.

**Consistency notions.** For a configuration $(\alpha, D)$, Willard distinguishes (2005, p. 5): *Level(0−)*: there is no $D$-proof of $0 = 1$ from $\alpha$; *Level(1)*: there is no $\Pi_1^*$ sentence $\Psi$ such that both $\Psi$ and $\neg\Psi$ have $D$-proofs from $\alpha$. "All the definitions of consistency, from Level(0-) up to Level(n) for any $n$, are equivalent to each other under strong enough models of arithmetic. However, many weak axiom systems do not have a mathematical strength to recognize this equivalence" (2005, p. 6, text layer). Willard's rule for choosing among them: "it is desirable to use the weakest possible definition ... when one is seeking to generalize the Second Incompleteness Theorem. On the other hand, the opposite is true when one seeks to develop 'boundary-case exceptions' " (2002c, p. 2, text layer). This is why his negative results below use Level(0−) and his positive ones Level(1).

**Provability, halting, and the Gödel sentence.** For each configuration in this essay, $\mathrm{Prf}_\alpha^D(x, y)$ ("$y$ is the Gödel number of a $D$-proof from $\alpha$ of the sentence with Gödel number $x$") is a bounded formula, and $\mathrm{Prov}_\alpha^D(x) := \exists y\, \mathrm{Prf}_\alpha^D(x, y)$. Oberhoff's «A(A) halts», for the proof-searching programs P, B, C of this essay, is formalized as "the search succeeds": an existential statement about Prf, not about execution traces. The sentence $\ddot{\mathsf{G}}$ is Willard's fixed point (2001, Eq. 63, p. 38):

$$\Gamma(g) := \forall x\, \forall h \le x\, \big[\mathrm{SUBST}(g, h) \supset \neg \mathrm{Prf}_\alpha^D(h, x)\big], \qquad \ddot{\mathsf{G}} := \Gamma(\bar{n}),$$

where $\mathrm{SUBST}(g, h)$ says that $h$ is the code of the sentence obtained by substituting the numeral of $g$ into the formula coded by $g$, and $n$ is the Gödel number of $\Gamma(g)$. $\ddot{\mathsf{G}}$ is a $\Pi_1$-class sentence. Let $c := \ulcorner \ddot{\mathsf{G}} \urcorner$.

Several sentences say that $\ddot{\mathsf{G}}$ is false, and they must be kept apart, because Prov is sensitive to codes. The *halting sentence*

$$\mathsf{H} := \exists x\, \exists h \le x\, [\mathrm{SUBST}(\bar{n}, h) \wedge \mathrm{Prf}_\alpha^D(h, x)]$$

is «P(P) halts», a $\Sigma_1$-class sentence. The *syntactic negation* $\neg\ddot{\mathsf{G}}$ is the string $\ddot{\mathsf{G}}$ with a negation sign in front. It belongs to neither class, since a $\Sigma_n^*$ sentence is one "written in the form $\exists v_1 \ldots \exists v_k\, \Phi$" (2005, p. 5), and Willard says of such a string that it "is neither $\Sigma_1^*$ nor $\Pi_1^*$ because a negation symbol appears to the left of its unbounded quantifiers" (p. 26, footnote 3, text layer). The two are logically equivalent, so every model satisfies $\neg\ddot{\mathsf{G}} \leftrightarrow \mathsf{H}$, and Fact A gives $\alpha \vdash_D \neg\ddot{\mathsf{G}} \leftrightarrow \mathsf{H}$. But they have different Gödel numbers, and $\alpha \vdash_D \neg\ddot{\mathsf{G}} \leftrightarrow \mathsf{H}$ does not give $\alpha \vdash_D \mathrm{Prov}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner) \leftrightarrow \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$. That would be a derivability condition, and derivability conditions are what this essay puts in question.

Willard's consistency axiom uses a bounded formula $\mathrm{Pair}(x, y)$ that holds when "$x$ is the Gödel number of a $\Pi_1^*$ sentence and $y$ is $x$'s negation" (2005, p. 9), "the sentence that represents the negation of $x$'s formal statement" (p. 34). At the level of codes his text admits two readings.

(i) $y$ codes the $\Sigma_1^*$ sentence that represents the negation. This is how his proof of Theorem 1 uses it: "$j_2$ will be a proof of the $\Sigma_1^*$ theorem that is $\Phi$'s negation" (p. 20). It is also how Remark 5, II (p. 27) treats a negated sentence, through its Prenex* equivalent.

(ii) $y$ codes the string $\neg\Phi$. This is how Eq. (28) writes it, $\mathrm{Pair}(\ulcorner f \urcorner, \ulcorner \neg f \urcorner)$ (p. 34), and how the Level($n$) definition writes "$\Upsilon$ and $\neg\Upsilon$" (p. 5).

We write $\mathsf{N}$ for the sentence that Pair pairs with $\ddot{\mathsf{G}}$, and $c' := \ulcorner \mathsf{N} \urcorner$. Under (i), $\mathsf{N}$ is a $\Sigma_1$-class sentence saying that the search succeeds: a formalization of «P(P) halts» on a par with $\mathsf{H}$, though not necessarily the same string. Under (ii), $\mathsf{N}$ is $\neg\ddot{\mathsf{G}}$. Either way $\alpha \vdash_D \mathsf{N} \leftrightarrow \neg\ddot{\mathsf{G}}$ by logic. Our results are stated for $\mathsf{N}$ and hold under either reading. Where the reading matters, we say so (§6, §7).

**Fact B** (Fixed point; Willard 2001, proof of Theorem 7.2, p. 39)**.** *Let $D$ be $\alpha$-sound and $\alpha$-complete, let $\mathrm{Prf}_\alpha^D$ be bounded, and let $\mathrm{SUBST}(\bar{n},\bar{c})$ be true. Suppose that $\alpha$ proves every true bounded sentence and proves trichotomy, $\forall y\,(y \le \bar{c} \vee \bar{c} \le y)$, and that the encoding satisfies (E): every $D$-proof of $\ddot{\mathsf{G}}$ has a Gödel number exceeding $c$. Then*

$$\alpha \vdash_D \ddot{\mathsf{G}} \supset \forall y\, \neg \mathrm{Prf}_\alpha^D(\bar{c}, y). \tag{FP$\Rightarrow$}$$

*If $\alpha$ also proves the uniqueness sentence (U) $\forall h\,[\mathrm{SUBST}(\bar{n},h) \supset h = \bar{c}]$, then*

$$\alpha \vdash_D \ddot{\mathsf{G}} \leftrightarrow \forall y\, \neg \mathrm{Prf}_\alpha^D(\bar{c}, y). \tag{FP}$$

*Proof.* The following are theorems of $\alpha$: trichotomy; the bounded sentence $\forall y \le \bar{c}\ \neg\mathrm{Prf}(\bar{c},y)$, which is true by (E) (it excludes $y = c$ as well); and the true bounded sentence $\mathrm{SUBST}(\bar{n},\bar{c})$. In every structure, their conjunction implies (FP$\Rightarrow$). Given $y$, either $y \le \bar{c}$, and the second theorem gives $\neg\mathrm{Prf}(\bar{c},y)$; or $\bar{c} \le y$, and $\ddot{\mathsf{G}}$ instantiated at $x = y$, $h = \bar{c}$ gives it, using the third. Adding (U), the conjunction also implies the converse: given $x$ and $h \le x$ with $\mathrm{SUBST}(\bar{n},h)$, (U) gives $h = \bar{c}$, and the hypothesis gives $\neg\mathrm{Prf}(\bar{c},x)$. So each implication holds in every model of $\alpha$, and Fact A, applied repeatedly to these theorems, gives its $D$-proof. □

Only the direction (FP$\Leftarrow$) needs (U). Willard's text gives (U) in two clauses: "$\Theta$'s Gödel number is the unique integer satisfying $\mathrm{SUBST}(\bar{n},\ulcorner\Theta\urcorner)$" and, of this uniqueness, "it is provable from PAX as a $\Pi_1^-$ theorem" (2001, p. 39, text layer). So (U) holds in any system that inherits the $\Pi_1^-$ theorems of PAX, Peano Arithmetic extended by the grounding functions. Trichotomy and (E) are our explicit additions.

(E) is an encoding hypothesis about proofs of the one $\Pi_1$-class sentence $\ddot{\mathsf{G}}$, and we checked it against Willard's encodings, under two assumptions that we state. In 2005, Appendix A (pp. 39–40), a proof is a string of six-bit bytes read as an integer in base 64. Each of the 24 symbols has a code between 32 and 55, so its byte has leading bit 1; the continuation bytes of a variable or parameter index have leading bit 0. A tableau proof of $\ddot{\mathsf{G}}$ opens with a tree-separating parenthesis, whose code is at least 32, followed by the root, which is $\neg\ddot{\mathsf{G}}$ rewritten in Prenex$^*$ form (p. 6), and it closes with a parenthesis. So a proof of $N$ bytes has value at least $32 \cdot 64^{N-1}$, and $N \ge |\mathrm{root}| + 2$.

*First assumption:* for a prenex sentence such as $\ddot{\mathsf{G}}$, Willard's Prenex$^*$ rewriting of the negation dualizes the unbounded quantifier prefix and negates the bounded matrix, possibly absorbing one leading negation, so that $|\mathrm{root}| \ge |\ddot{\mathsf{G}}| - 1$. For non-prenex sentences the rewriting can shorten much more, for instance by cancelling stacked double negations, and the argument below does not apply to them. *Second assumption:* a sentence's Gödel number is the value of its byte string. 2001 states this explicitly (Appendix B, p. 51); for 2005 we assume the same.

Under the two assumptions, $c = \ulcorner\ddot{\mathsf{G}}\urcorner < 64^{|\ddot{\mathsf{G}}|} \le 32{\cdot}64^{|\ddot{\mathsf{G}}|} \le 32{\cdot}64^{N-1}$. A Tab$-U_1^*-$List proof of $T$ is a list of tableau proofs $p_1,\ldots,p_n$ with $T$ the conclusion of $p_n$ (p. 7), encoded as the concatenation of their byte strings (p. 40). So the list's value is at least $p_n$'s, and (E) for $p_n$ gives (E) for the list. 2001's encoding (Appendix B, pp. 51–52) is the same scheme, with every symbol byte led by bit 1 and each tableau node opened by a curly bracket, so the same argument applies.

(E) compares a proof with its *conclusion*. Willard's construction also uses a companion fact, that an axiom used in a proof has a code no larger than the proof's: the bound $h \le p$ in his Eq. (7) (2005, p. 10). The displayed formula imposes that bound, and the prose asserts its equivalence with the unbounded existential. We take that from Willard and do not derive it from (E).

**Relative consistency.** Every consistency result in this essay, Willard's and ours, is a theorem of ordinary metamathematics proved under a hypothesis, typically that the $\Pi_1$-class theorems of a

base theory $A$ are true. That hypothesis is not decidable, and no system here is "proved consistent" outright. We have not determined how much of the metatheory Willard's proofs use.

## 3 Where the Faith Lives

Here is the first half of Oberhoff's proof that $\ddot{\mathsf{G}}$ is undecidable (his §4, a second proof of his Theorem 4):

**Suppose $\ddot{\mathsf{G}}$ is provable:**
**Or $\neg\ddot{\mathsf{G}}$ isn't provable:**

⇒ Because $\ddot{\mathsf{G}}$ is provable $\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$.

⇒ Because this sequence leads to a terminal state this gives a proof of «$\mathsf{P}(\mathsf{P})$ halts» = $\neg\ddot{\mathsf{G}}$. ↯

Theorem 6 asks $\mathcal{F}$ to run this argument on itself. Fix a configuration $(\mathcal{F}, D)$ and write Prov for $\mathrm{Prov}^{D}_{\mathcal{F}}$. As statements that $\mathcal{F}$ must prove, the argument has three steps.

(1) $\neg\ddot{\mathsf{G}} \supset \mathrm{Prov}(\bar{c})$: «if $\mathsf{P}(\mathsf{P})$ halts, then $\ddot{\mathsf{G}}$ is provable». This is (FP⇐), contraposed. It is not literally trivial: $\neg\ddot{\mathsf{G}}$ is equivalent to $\mathsf{H}$, which speaks of "the sentence $h$ with $\mathrm{SUBST}(\bar{n}, h)$", and identifying $h$ with $\ddot{\mathsf{G}}$ needs the uniqueness sentence (U). With (U), it is cheap.

(2) $\mathrm{Prov}(\bar{c}) \supset \mathrm{Prov}(\bar{c}\,')$: «if $\ddot{\mathsf{G}}$ is provable, then its negation $\mathsf{N}$ is provable». This is the step "$\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$". Inside $\mathcal{F}$ it reads: *for every proof $s$ of $\ddot{\mathsf{G}}$ there is a proof of $\mathsf{N}$*, namely the proof that exhibits $s$ and checks it, so proving that the search succeeds, written as $\mathsf{N}$. We aim the step at $\mathsf{N}$ because step (3) needs the code that the consistency statement pairs with $c$. A variant aimed at another formalization of the halting sentence, such as $\mathrm{Prov}(\bar{c}) \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$ when $\mathsf{H}$ differs from $\mathsf{N}$, yields step (2) only together with the formalized conversion $\mathrm{Prov}(\ulcorner\mathsf{H}\urcorner) \supset \mathrm{Prov}(\bar{c}\,')$.

(3) $\mathrm{Prov}(\bar{c}) \wedge \mathrm{Prov}(\bar{c}\,') \supset \mathit{Inc}$, where $\mathit{Inc}$ is the negation of the consistency statement in use. For Level(1), whose statement is $\forall x\,\forall y\,\forall p\,\forall q\ \neg[\mathrm{Pair}(x,y) \wedge \mathrm{Prf}(x,p) \wedge \mathrm{Prf}(y,q)]$ with Pair as in §2 (2005, Eq. 6, p. 9), step (3) follows from this statement and the true bounded sentence $\mathrm{Pair}(\bar{c}, \bar{c}\,')$ by Fact A. For Level(0−) it is a formalized combination of two proofs into a proof of $0 = 1$, which for a cut-free $D$ is the expensive construction Fact A warns about.

Chaining (1)–(3) gives $\mathcal{F} \vdash_D \neg\ddot{\mathsf{G}} \supset \mathit{Inc}$, that is, $\mathcal{F} \vdash_D \neg\mathit{Inc} \supset \ddot{\mathsf{G}}$: «$\mathcal{F}$ is consistent ⇒ $\ddot{\mathsf{G}}$», which is what Theorem 6 needs.

The only step that asks $\mathcal{F}$ to know something substantial about itself is (2). In the first incompleteness theorem, "which we can write out in $\mathcal{F}$" was a fact *about* $\mathcal{F}$, established by us from outside: we are handed a halting run and we produce the proof. In the second theorem it has to be a theorem *of* $\mathcal{F}$, stated for every $s$ at once. The two deserve separate names.

**Definition C** (Watching)**.** $(\mathcal{F}, D)$ *watches* if $\mathcal{F} \vdash_D \sigma$ for every true $\Sigma_1$-class sentence $\sigma$ of its language.

**Definition D** (Knowing that one watches)**.** $(\mathcal{F}, D)$ *knows that it watches* at a sentence $\sigma$ if $\mathcal{F} \vdash_D \sigma \supset \mathrm{Prov}(\ulcorner\sigma\urcorner)$. It *knows that it watches* if it does so at every $\Sigma_1$-class sentence and at the syntactic negation of every $\Pi_1$-class sentence.

A program halts if and only if a $\Sigma_1$ sentence about it is true, so watching is external $\Sigma_1$-completeness, and knowing that one watches is formalized $\Sigma_1$-completeness. The second clause of the definition is ours. It is there because Prov is sensitive to codes, and under reading (ii) of Pair the sentence that matters, $\mathsf{N} = \neg\ddot{\mathsf{G}}$, is a syntactic negation. Where the derivability conditions hold, each written form of a $\Sigma_1$ fact gives the other; for Willard's systems this is not known. The third Hilbert–Bernays–Löb derivability condition [Hilbert–Bernays 1939, Löb 1955], $\mathrm{Prov}(\ulcorner\varphi\urcorner) \supset \mathrm{Prov}(\ulcorner\mathrm{Prov}(\ulcorner\varphi\urcorner)\urcorner)$, is the instance at the $\Sigma_1$ sentences $\sigma = \mathrm{Prov}(\ulcorner\varphi\urcorner)$, the halting statements of proof searches.

**Lemma E** (ours)**.** *Let $(\mathcal{F}, D)$ watch, with $D$ $\mathcal{F}$-sound and $\mathcal{F}$-complete and $\mathrm{Prf}^D_{\mathcal{F}}$ correct in the standard model (it holds of $(h, x)$ exactly when $x$ codes a $D$-proof from $\mathcal{F}$ of the sentence coded $h$).*

*(a) $(\mathcal{F}, D)$ knows that it watches at every* true *sentence that is $\Sigma_1$-class or the syntactic negation of a $\Pi_1$-class sentence.*

*(b) If (FP) holds, step (2) is $\mathcal{F}$-provably equivalent to knowing that one watches at $\mathsf{N}$: $\mathcal{F} \vdash_D (2) \leftrightarrow [\mathsf{N} \supset \mathrm{Prov}(\bar{c}\,')]$.*

*Proof.* (a) Let $\sigma$ be true. Then $\mathcal{F} \vdash_D \sigma$: by watching if $\sigma$ is $\Sigma_1$-class, and if $\sigma$ is $\neg\pi$ with $\pi$ $\Pi_1$-class, because $\sigma$ is logically equivalent to a true $\Sigma_1$-class sentence, which $\mathcal{F}$ proves, and Fact A applies. So a $D$-proof of $\sigma$ exists, $\mathrm{Prov}(\ulcorner\sigma\urcorner)$ is a true $\Sigma_1$-class sentence, and watching gives $\mathcal{F} \vdash_D \mathrm{Prov}(\ulcorner\sigma\urcorner)$. Since $\mathrm{Prov}(\ulcorner\sigma\urcorner) \supset (\sigma \supset \mathrm{Prov}(\ulcorner\sigma\urcorner))$ is valid, Fact A gives the implication. (b) (FP) and $\mathsf{N} \leftrightarrow \neg\ddot{\mathsf{G}}$ are theorems of $\mathcal{F}$, so by soundness every model of $\mathcal{F}$ satisfies $\mathsf{N} \leftrightarrow \mathrm{Prov}(\bar{c})$. The two implications differ only by this replacement in the antecedent, so their equivalence holds in every model of $\mathcal{F}$, and Fact A gives the $D$-proof. □

So the faith step is not a claim about runs that happen. Those a watching system sees, and Lemma E(a) gives it the knowledge for free. The faith is a claim about a run that does not happen: if $\ddot{\mathsf{G}}$ is true, $\mathsf{H}$, $\neg\ddot{\mathsf{G}}$ and $\mathsf{N}$ are false, and step (2) asserts that *had* $\mathsf{P}(\mathsf{P})$ halted, $\mathcal{F}$ would have seen it.

Here is what each result of [Oberhoff 2018] uses:

| Result in [Oberhoff 2018] | Assumes about $\mathcal{F}$ | Watches | Knows it watches |
|---|---|---|---|
| Thm 4 (G1 via halting) | honest; complete (for contradiction) | — | — |
| §4, $\ddot{\mathsf{G}}$ undecidable | consistent; honest in case 2 | case 1 | — |
| Thm 5 (Rosser, via $\mathsf{B}$) | consistent | both cases | — |
| Thm 6 (G2) | consistent | yes | step (2) |

Theorem 4 needs only that $\mathcal{F}$ can *state* halting facts honestly; the search does the rest. Case 2 of the §4 argument uses only that $\mathcal{F}$'s $\Sigma_1$ theorems are true. Rosser's case 2 also uses a $\Pi_1$ fact, that a run which has entered `while true {}` never leaves it; Oberhoff's footnote 12: "Any formal system capable of reasoning about algorithms can prove that 'while true {}' runs forever; otherwise it ought to be ashamed of itself." Apart from that, every result in the essay except the second incompleteness theorem uses watching and nothing more. The classical proof of the second theorem runs through the derivability conditions; the computational retelling shows what the third condition *says*.

## 4 Transcripts Are Bigger Than Runs

Why should knowing be harder than watching? The natural proof of step (2) takes a proof $s$ of $\ddot{\mathsf{G}}$ and produces the proof $t$ of «$\mathsf{P}(\mathsf{P})$ halts» that exhibits $s$ and checks it. In a language whose only growth functions are $+$ and *Double*, exhibiting $s$ costs a term, and terms are expensive.

**Lemma F** (ours)**.** *In Willard's 2005 language and encoding, every semantic tableau proof that contains a closed term of value $v \geq 2$ has Gödel number greater than $v^5$.*

*Proof.* A closed term with $L$ function symbols built from $\bar{c}_0$, $\bar{c}_1$ has value at most $2^L$: non-growth functions do not increase the maximum of their arguments, and $+$ and *Double* at most double it. This is the doubling estimate of case (c) in Willard's proof of Lemma 1 (2005, p. 24), applied to a term with no parameters. So a term of value $v$ has $L \geq \log_2 v$ function symbols. The U-Length of a node is the number of U-Grounding function symbols in its sentence; its U-Depth is its own U-Length plus the U-Lengths of all its ancestors; and the U-Height of the tree is the largest U-Depth of a leaf (p. 22). So the node containing the term has U-Depth at least $L$. If that node is a leaf, the U-Height is at least its U-Depth. If not, a leaf below it has a U-Depth at least as large, since the nodes in between add non-negative U-Lengths. Either way the tree has U-Height at least $L$. Willard's Eq. (20) (p. 26) is U-Height$(p) < \frac{1}{5}\log_2 p$, which, he says, Appendix A's encoding "as well as any other reasonable encoding methodology, will clearly satisfy" (text layer). Hence $\log_2 p > 5L \geq 5\log_2 v$. □

So the transcript $t$ of step (2) is bigger than the fifth power of the run $s$ it records, and the natural proof of step (2) asks $\mathcal{F}$ to believe that $s \mapsto t$ is defined everywhere. A totality claim for a function that grows like a power is a claim about multiplication, which these systems do not make. This is a heuristic, not a theorem: step (2) is vacuously true when $\ddot{\mathsf{G}}$ is, and a system could in principle prove it without exhibiting anything. Section 7 closes that loophole by a different route, and Section 8 shows a bounded transcript statement at work in a theorem of Willard's.

Willard's first statement of the idea continues this essay's epigraph (TR 93-10, p. 7, verbatim):

> This means there will always be adequate memory for writing $b_n$'s binary encoding when there is sufficient memory for storing a semantic tableaux proof of its existence (whereas the same is plainly not true *when multiplication is assumed to be a function*).
>
> For instance, multiplication allows one to establish the existence of a number $a_{300}$ (in a proof with approximately 300 lines), whose binary encoding requires *more digits* than the number of atoms in the universe.

Here $b_{i+1} = b_i + b_i$ and $a_{i+1} = (a_i)^2$. Starting both at 2, $b_n = 2^{n+1}$ and $a_n = 2^{2^n}$: a proof of $n$ doublings names a number of $n+1$ bits, and a proof of $n$ squarings names one of $2^n$ bits. Willard's later statement compares the Gödel number of a proof with the integer it shows to exist (2006a, Remark 3, p. 24, verbatim):

> In particular, every proof of the existence of an integer $N$ using the additive convention will have a larger Gödel number than $N$. In contrast, the multiplicative convention allows for the Gödel number of such proofs to have a sharply smaller magnitude than $N$ (when for instance $N = 2^{2^k}$ and $k$ is larger than some fixed constant). In essence, this difference in magnitude is the reason the Second Incompleteness Theorem will generalize under the multiplicative naming paradigm — while Section 4 showed boundary-case exceptions to it exist under the additive naming convention.

## 5 A Machine That Says "I Am Consistent"

Oberhoff's moral for the second theorem is "if you declare yourself a very stable genius, you're not". Let us build the genius with his tools. Fix a base system $\mathcal{F}_0$, a deduction method $D$, and a consistency level, and consider

```
C(A) {
  for s ∈ all possible strings {
    if s is a contradiction in F_0 + «A(A) doesn't halt» {
      halt;
    }
  }
}
```

where a *contradiction* is a $D$-proof of $0 = 1$ (Level(0−)) or a pair of $D$-proofs of a $\Pi_1$-class sentence and of its syntactic negation (Level(1)). (We picked C because it is a circle that never quite closes; §6 explains the joke.) Let

$$\ddot{\mathsf{W}} = \text{«C(C) doesn't halt»}, \qquad \mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}.$$

Just as $\ddot{\mathsf{G}}$ says "I am not provable", $\ddot{\mathsf{W}}$ says that $\mathcal{F}$ is consistent at the chosen level. So $\mathcal{F}$ proves its own consistency in one line: the claim is an axiom.

C is not an analogy. It is Willard's construction read as a program. In 2005 (pp. 9–10), $ExPrf^{D}_{\mathrm{UNION}(A)}(h,t,p)$ says that $p$ is a $D$-proof of $t$ from Groups 0–2 plus the sentence coded $h$, and $SubstPrf^{D}_{\mathrm{UNION}(A)}(g,t,p)$, his Eq. (7), combines it with $\mathrm{SUBST}(g,h)$: the input A is $g$, and «A(A) doesn't halt» is the sentence $h$ with $\mathrm{SUBST}(g,h)$. His Group-3 axiom, Eq. (8), is $\Gamma_3(\bar{n}_3)$, where $n_3$ is the Gödel number of

$$\Gamma_3(g) := \forall x\, \forall y\, \forall p\, \forall q\, \neg[\mathrm{Pair}(x,y) \wedge SubstPrf^{D}_{\mathrm{UNION}(A)}(g,x,p) \wedge SubstPrf^{D}_{\mathrm{UNION}(A)}(g,y,q)]$$

(Willard writes $\Gamma$ and $n$; we add the subscript to keep it apart from the Gödel sentence's $\Gamma$ and $n$ of §2). This is «C(C) doesn't halt» at Level(1); 2001's Appendix B does the same at Level(0−). Willard (2016, Definition 3.4 and Example 3.5) remarks that constructing such a sentence is "relatively easy, via an application of the Fixed Point Theorem, but this sentence is ironically typically useless!"

**Proposition G** (The genius falls; standard)**.** *Let $\mathcal{F}_0$ be a recursively axiomatized extension of $I\Sigma_1$ in the language of arithmetic, D Hilbert-style deduction,* Prov *the canonical $\Sigma_1$ provability predicate, and* $\ddot{\mathsf{W}}$ *as above at either level. Then $\mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}$ is inconsistent, and* C(C) *halts.*

*Proof.* Suppose $\mathcal{F}$ is consistent.

- ⇒ $\mathcal{F}$ proves its own consistency: $\ddot{\mathsf{W}}$ is one of its axioms, and by the diagonal lemma $I\Sigma_1$ proves $\ddot{\mathsf{W}}$ equivalent to the consistency statement of $\mathcal{F}_0 + \ddot{\mathsf{W}}$ at the chosen level, for the canonical proof predicate. $I\Sigma_1$ also proves the two levels equivalent for $\mathcal{F}$: it proves explosion for the canonical Hilbert proof predicate, so a proof of $0 = 1$ yields proofs of every sentence, and it proves that proofs of a sentence and of its negation combine, by modus ponens, into a proof of $0 = 1$.
- ⇒ $\mathcal{F}$ is a recursively axiomatized extension of $I\Sigma_1$, so the derivability conditions hold for its canonical provability predicate, provably in $I\Sigma_1$ [Hájek–Pudlák 1993]. In Oberhoff's terms, $\mathcal{F}$ knows that it watches, and steps (1)–(3) are theorems of $\mathcal{F}$.
- ⇒ So by the second incompleteness theorem (Oberhoff's Theorem 6, with its faith step now a theorem), $\mathcal{F}$ cannot prove its own consistency. ↯

So $\mathcal{F}$ is inconsistent. A contradiction from $\mathcal{F}$ is then a finite object that exists, and C(C) finds it. □

Willard records that "Kleene, Rogers and Jeroslow [...] each warned their readers that most axiom systems, similar to $\alpha^*$, were useless on account of their inconsistency, although they were

technically well-defined" (2005, §1). Two of his own theorems are instances in his own languages. XIS$^{\lambda}$(PAX), his tableau system with multiplication total and a reflection-style Group-3 axiom, "is inconsistent" for every $\lambda$ (2001, Theorem 7.3, p. 40, `full`). And Section 8 describes the finest instance, where the system falls because of one added axiom saying that one number exists.

# 6 Willard's Machine

Willard keeps $\mathsf{C}$ and $\ddot{\mathsf{W}}$ and changes $\mathcal{F}_0$ and $D$. We describe the 2005 system $\mathrm{IS}_D(A)$ and note where the 2001 system $\mathrm{IS}(A)$ differs.

**The system $\mathrm{IS}_D(A)$ (2005, §3, pp. 8–10).** $A$ is any axiom system in the U-Grounding language whose proof predicate $\mathrm{Prf}_A$ is $\Delta_0^*$, and $D$ is a deduction method.

- **Group-0** (names): four axioms fixing $\bar{c}_0$, $\bar{c}_1$, $+$ and *Double*, so that every $n \geq 2$ has the name $\overline{n}$ of $O(\log n)$ symbols.
- **Group-1** (watching): a finite set of true $\Pi_1^*$ sentences which, with Group-0, proves every $\Delta_0^*$ sentence true in the standard model. Willard's Table I (2001, pp. 63–64) is one such set.
- **Group-2** (inheritance): for each $\Pi_1^*$ sentence $\Phi$, the axiom $\forall p\,\{\mathrm{Prf}_A(\ulcorner\Phi\urcorner, p) \rightarrow \Phi\}$ (2005, Eq. 5). The schema is indexed by all $\Pi_1^*$ sentences, not by $A$'s theorems; for $\Phi$ that $A$ does not prove, the axiom is vacuously true.
- **Group-3** (self-reference): the single $\Pi_1^*$ sentence $\Gamma_3(\bar{n}_3)$ of §5, Level(1) consistency of $\mathrm{IS}_D(A)$ under $D$, with $\mathrm{Prf}^D_{\mathrm{IS}_D(A)}(t,p)$ encoded as $\mathit{SubstPrf}^D_{\mathrm{UNION}(A)}(\bar{n}_3, t, p)$ (p. 10). Willard stresses that the $\Pi_1^*$ form is essential (2005, p. 10).

The deduction methods are *semantic tableaux* (2005, §2, pp. 6–7: a tree rooted at $\neg\Phi$ in prenex form, with elimination rules only and no cut, every branch closed by some $\Upsilon$ and $\neg\Upsilon$) and Tab$-U_1^*-$List: a list of tableau proofs, each of which may use the earlier conclusions as axioms, all intermediate conclusions being $\Pi_1^*$ or $\Sigma_1^*$ (2005, p. 7). The latter is a restricted modus ponens: lemmas are allowed, provided each has at most one unbounded quantifier block.

$\mathrm{IS}(A)$ of 2001 differs in three respects: its language is the 2001 grounding language, with a constant for every number and addition total by an axiom rather than a function symbol; $D$ is semantic tableaux; and its Group-3 axiom asserts Level$(0-)$ consistency, $\forall y\,\neg \mathit{SemPrf}_{\mathrm{IS}(A)}(\ulcorner 0 = 1\urcorner, y)$ (2001, Eq. 8), with a $\Pi_1^-$ encoding (2001, Theorem C.3, p. 60). Willard assumes throughout that $A$ extends Peano Arithmetic (2001, p. 9).

**Theorem H** (Willard)**.**

*(a) (2005, Theorems 3 and 5, pp. 27, 29, `full`.) If every $\Pi_1^*$ theorem of $A$ is true in the standard model, then $\mathrm{IS}_D(A)$ is consistent, both for $D$ = semantic tableaux and for $D = \mathrm{Tab}-U_1^*-\mathrm{List}$. The proof (via Theorem 1) rules out $D$-proofs of a $\Pi_1^*$ sentence and of "the $\Sigma_1^*$ theorem that is $\Phi$'s negation" (p. 20), so the consistency is Level(1), with negation in the sense of reading (i) of §2.*

*(b) (2001, Theorem 4.3, p. 20, `full`.) If $A$ is* regularly consistent*, meaning that $\mathrm{Prf}_A$ has a $\Delta_0^-$ encoding and every Group-2 axiom of $\mathrm{IS}(A)$ is true (2001, p. 12), then $\mathrm{IS}(A)$ is consistent: there is no tableau proof of $0 = 1$, which is Level$(0-)$.*

Both are relative consistency theorems in the sense of §2. Willard also states that IS's construction fails for Hilbert deduction: by Pudlák and Solovay, "if $D$ denotes Hilbert deduction then our $\mathrm{IS}_D(A)$ system will be automatically inconsistent" (2005, p. 8, text layer).

**Lemma I** (ours)**.** *For $D$ = semantic tableaux or* Tab$-U_1^*-$List*,* $(\mathrm{IS}_D(A), D)$ *watches.*

*Proof.* Let $\sigma = \exists\vec{x}\,\delta(\vec{x})$ be a true $\Sigma_1^*$ sentence with witnesses $\vec{k}$. Then $\delta(\bar{\vec{k}})$ is a true $\Delta_0^*$ sentence, which Groups 0 and 1 prove, and $\sigma$ is its logical consequence. Apply Fact A. □

The same holds for $\mathrm{IS}(A)$ with $A \supseteq$ PAX: a system that verifies the $\Pi_1^-$ theorems of PAX "obviously has a capacity to prove every $\Delta_0^-$ sentence which is true" (2001, p. 39, text layer), and $\mathrm{IS}(A)$ names every number by a constant. So a true $\Sigma_1^-$ sentence $\exists\vec{x}\,\delta(\vec{x})$, with witnesses $\vec{k}$, has the true $\Delta_0^-$ instance $\delta(\bar{\vec{k}})$, which $\mathrm{IS}(A)$ proves, and Fact A adds the existential quantifiers, as in Lemma I. For general programs, Willard notes that the predicate $\mathit{Turing}(x, y, z)$ is expressible in $\mathrm{IS}(A)$ (TR 93-10, p. 6), so halting of any program is a $\Sigma_1^-$ sentence there; the arguments below need only the proof-searchers, whose halting is formalized through Prf directly.

**Lemma J** (ours)**.** *Let every $\Pi_1^*$ theorem of $A$ be true, and read* Pair *as in (i) of §2. Then every theorem of* $\mathrm{IS}_D(A)$ *is true in the standard model. Under reading (ii) the same holds given the Prenex$^*$ convention discussed after Theorem 1's proof below.*

*Proof.* Groups 0 and 1 are true (Willard: "It is obvious that all $\mathrm{IS}_D(A)$'s Group-0 and 1 axioms are valid under the standard model", 2005, p. 20, text layer). Group-2 is true by hypothesis. Group-3 is true because Theorem H(a) rules out proofs of a $\Pi_1^*$ sentence and of its $\Sigma_1^*$ negation, and under reading (i) that is what Group-3 says, its proof predicate $\mathit{SubstPrf}^D_{\mathrm{UNION}(A)}(\bar{n}_3, \cdot, \cdot)$ being correct in the standard model by Willard's construction (2005, pp. 9–10). Theorems of true axioms under a sound $D$ are true. □

So $\mathrm{IS}_D(A)$ is honest in Oberhoff's sense, under the hypotheses of Lemma J. If in addition $A$ proves the $\Pi_1^*$ theorems of PA+ (Peano Arithmetic with the U-Grounding function symbols, 2005, p. 32), it obeys the first incompleteness theorem: by Oberhoff's §4 argument (spelled out as Proposition K(b) and Corollary L below), $\ddot{\mathsf{G}}$ is unprovable, and $\neg\ddot{\mathsf{G}}$ is false, hence unprovable by honesty. Rosser's version also goes through. In both cases the finite search that Oberhoff writes out is a true bounded fact, which Groups 0 and 1 prove. In case 2, $\mathsf{B}(\mathsf{B})$ enters its loop at a definite step, and PA+ proves the $\Pi_1^*$ sentence $\Phi =$ «$\mathsf{B}(\mathsf{B})$ doesn't halt». Group-2 hands it to $\mathrm{IS}_D(A)$ by detachment: $A$ proves $\Phi$ with some proof $p$, Groups 0 and 1 prove the true $\Delta_0^*$ sentence $\mathrm{Prf}_A(\ulcorner\Phi\urcorner, \bar{p})$, and with $\Phi$'s Group-2 axiom Fact A gives $\Phi$. Willard said so in his first technical report: $\mathrm{IS}(A)$ is "a class of structures, lying halfway between Peano Arithmetic and Presburger Arithmetic, which satisfy Godel's First but not Second Incompleteness Theorem" (TR 93-10, p. 6, verbatim).

**Why it is consistent, in Oberhoff's format.** Willard's 2005 proof needs two definitions. A $\Pi_1^*$ sentence $\forall\vec{x}\,\psi$ is *App*$\forall(a)$*-valid* if $\psi$ holds for all $\vec{x} \le a$; a $\Sigma_1^*$ sentence $\exists\vec{x}\,\psi$ is *App*$\exists(b)$*-valid* if it has a witness $\vec{x} \le b$ (Eqs. 16–17; bounded quantifiers are left alone). For $a > b \ge 1$, $\alpha$ is *Normed*$(a, b)$ if every axiom is a $\Pi_1^*$ sentence with App$\forall(a)$ validity or a $\Sigma_1^*$ sentence with App$\exists(b)$ validity, in the U-Grounding language with no other function symbol (Definition 4, p. 19). $D$ is *$\theta$-compactified*, for a constant $0 < \theta < 1/3$, if whenever $p \le \lceil a/b \rceil + 1$ is a $D$-proof of $\Phi$ from a Normed$(a, b)$ system, a $\Sigma_1^*$ $\Phi$ is App$\exists(b \cdot p^\theta)$-valid and a $\Pi_1^*$ $\Phi$ is App$\forall(a/p^\theta)$-valid (Definition 5, p. 19). In words: a proof $p$ can push an existential witness out by at most a factor $p^\theta$, and can shrink a universal's certified range by at most the same factor. Tableaux are $\frac{1}{4}$-compactified (Theorem 2, p. 26, `full`), and so are tableau lists (Theorem 4, p. 28, `full`). Here is Theorem 1 (p. 19, `full`), with $\theta = \frac{1}{4}$:

**Suppose $\mathrm{IS}_D(A)$ is inconsistent:**

- $\Rightarrow$ Then $\mathsf{C}(\mathsf{C})$ halts: there are $D$-proofs $j_1$ of a $\Pi_1^*$ sentence $\Phi$ and $j_2$ of the $\Sigma_1^*$ sentence $\Psi$ that is $\Phi$'s negation (Willard's wording, p. 20). Choose them with $m = \max(j_1, j_2)$ as small as possible.
- $\Rightarrow$ Groups 0–2 are true. Group-3 is false, but its first counterexample is a pair of proofs whose maximum is $m$, so it is App$\forall(m-1)$-valid. Hence $\mathrm{IS}_D(A)$ is Normed$(m-1, 1)$, which presupposes $m - 1 > 1$; under Willard's encoding every proof has Gödel number far above 3.
- $\Rightarrow$ The proof $j_1 \leq m$ certifies the universal $\Phi$ out to $(m-1)/m^{1/4}$. The proof $j_2 \leq m$ finds a witness for the existential $\Psi$ at most $m^{1/4}$.
- $\Rightarrow$ Since $(m-1)/m^{1/4} > m^{1/4}$, the witness for $\Psi$ lies where $\Phi$ has been certified. $\lightning$

The work is in the third arrow, $\theta$-compactification, and it is a program. For Part A of Theorem 2 (p. 26), suppose a tableau $p$ proves a $\Sigma_1^*$ sentence $\Phi$ from a Normed$(a, b)$ system but $\Phi$ has no witness $\leq b\,p^{1/4}$:

**Suppose $p$ is a tableau proof of $\Phi$ and $\Phi$ is not App$\exists(b\,p^{1/4})$-valid:**

- $\Rightarrow$ Then the root $\neg\Phi$ is App$\forall(b\,p^{1/4})$-valid, and the axioms together with the root form a Normed$(b\,p^{1/4}, b)$ system.
- $\Rightarrow$ Run *Probe* (Lemma 1, p. 23, `full`): walk down $p$ from the root; at a fork take a child that still satisfies the branch condition (each universal valid up to $b\,p^{1/4}$, each existential witnessed below $b$, each bounded sentence true) under the current valuation; when an existential is eliminated, give its new parameter the least value that keeps the branch true. Every parameter introduced at a node of U-Depth $\Delta$ gets a value at most $b \cdot 2^{\Delta}$ (Eq. 19), because terms grow at most by doubling.
- $\Rightarrow$ The walk succeeds if the tree's U-Height is below $\log_2(b\,p^{1/4}) - \log_2 b = \frac{1}{4}\log_2 p$, which Eq. (20) guarantees.
- $\Rightarrow$ A branch found by *Probe* cannot close (Lemma 2, p. 25, `full`): a closing pair must be bounded sentences (footnote 3), and both would be true under the valuation.
- $\Rightarrow$ So $p$ has an open branch, and is not a proof. $\lightning$

Part B, for $\Pi_1^*$ sentences, permutes the quantifiers (Appendix B, pp. 40–41). Theorem 3, for tableaux, is then immediate, and Theorem 5, for tableau lists, follows the same way from Theorem 4.

*The reading of* Pair. Theorem 1 rules out proofs of $\Phi$ and of its $\Sigma_1^*$ negation $\Psi$, because App$\exists$ validity is defined for $\Sigma_1^*$ sentences (Definition 5). That is the truth of Group-3 under reading (i). Under reading (ii), Group-3 speaks of proofs of the string $\neg\Phi$, and Theorems 3 and 5 give its truth only together with a passage from a proof of $\neg\Phi$ to a proof of $\Psi$. A tableau proof of $\neg\Phi$ has root $\neg\neg\Phi$ in Prenex$^*$ form, and a tableau proof of $\Psi$ has root $\neg\Psi$ in Prenex$^*$ form. Under a Prenex$^*$ convention that sends both to the same sentence, one tree is a proof of both and the passage is free. Willard's p. 6 does not fix the convention, and his "If $\mathrm{IS}_D(A)$ is inconsistent, then such a triple $(j_1, j_2, m)$ will clearly exist" (p. 20, text layer) passes over the point. We read Theorem H(a) under (i), or under (ii) with such a convention.

Willard's earlier intuition for the 2001 system, his "Assertion ++", is the same horizon argument seen from the other side (2001, Remark 4.4, p. 20, verbatim): "any proof $y$ of $0 = 1$ from IS$(A)$ will be automatically associated with an ordered pair $(p, u)$ where $p \leq \mathrm{VALUE}(u) < y$ and $p$ represents ANOTHER proof of $0 = 1$". He calls this a *vacuous truth*, an implication whose hypothesis is never

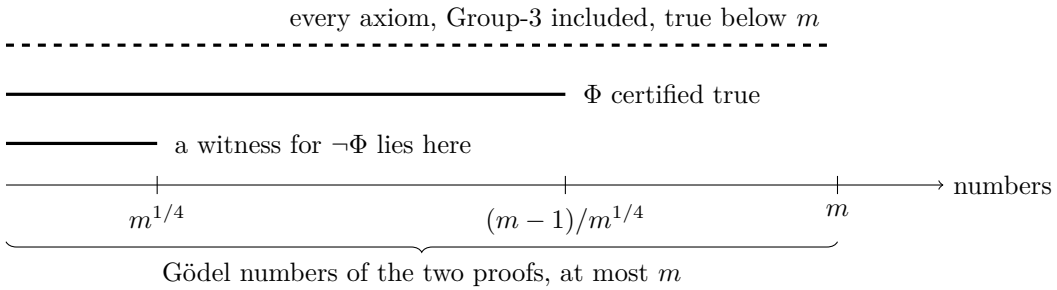



Figure 1: The two horizons in the proof of Willard's 2005 Theorem 1 (schematic, not to scale). A universal theorem is certified out to its horizon, an existential one finds its witness inside its horizon, and when horizons are short compared with the proofs, the two collide in the region where every axiom is true.

satisfied. The 2001 proof replaces it by an eleven-case induction showing that the least tableau proof $y$ of $0 = 1$ contains a branch on which every sentence is true once its unbounded quantifiers are cut off at $y - 1$ (Lemma 4.8, p. 23, `full`), which cannot close (Lemma 4.6, p. 22, `full`). The Group-3 axiom is false, but its cut-off at $y - 1$ is true, because $y$ is the least refutation. The naming bounds are Lemma 4.1 (a constant mentioned in $y$ has value below $\sqrt{y}$, p. 17, `full`) and Lemma 4.2 (along such a branch every parameter has value below $y$, p. 19, `sketch`), whose intuition is that fewer than $\frac{1}{3}\log_2 y$ doublings take values at most to $y^{1/2} \cdot y^{1/3} = y^{5/6} < y$ (the product is our unpacking of the sketch). The formal proof uses a different majorant, Part II of the affirmative-branch definition, $\mathrm{VALUE}(u) < (k + \sqrt{y}) \cdot 2^{\lfloor \beta_i/6 \rfloor - 2}$ (Eq. 26, p. 22), where $\beta_i$ is the bit-length of the first $i$ levels of the branch and each doubling is paid for by the six bits of Case 11. Willard later found this proof "hard to visualize" and replaced it by Theorem 1 (2005, Remark 2, p. 21).

Three remarks.

*The proof is a program. Probe* takes any purported proof and returns an open branch together with values for its parameters. Willard's consistency proof is therefore a machine that defeats each refutation by running it, in a universe barely larger than the refutation itself. That is self-justification ex machina in the most literal sense. (And it is why we named the searcher C: it hunts for a tree all of whose branches close, and the walk always finds one that stays open.) $\mathcal{F}$ does not prove that the machine works; that is the external part. $\mathcal{F}$ has only the axiom.

*The encoding enters as a floor.* The horizon is measured in Gödel numbers, so the argument needs every symbol to cost bits: Eq. (20) requires on average more than five bits per U-Grounding function symbol along the deepest branch, and 2001's Case 11 needs the sentence introducing a sum to require "substantially more than six bits" (2001, p. 28). Larger Gödel numbers only widen the margin, and Willard says so: "All our theorems would also be valid for uncompressed encoding methods. However, they are more meaningful when one uses an efficiently compressed B-adic encoding method" (2005, p. 40, text layer). 2006a makes the analogous remark for the Hilbert line: its theorems "have analogs under such uncompressed encoding methods, but they are substantially more meaningful" when compressed (Appendix A, p. 37, text layer). What would break the 2005 proof of Theorem 2 is a notation *more* compact than its argument allows. Part A needs $\text{U-Height}(p) < \frac{1}{4}\log_2 p$ (footnote 5, p. 27), so the third arrow fails once there are on average four or fewer bits per U-Grounding function symbol along the deepest branch. Whether Theorem 3 itself fails for such an encoding Willard does not say, and we do not know.

*The closing inequality.* Willard closes the proof of Theorem 1 with "since $\theta < \frac{1}{3}$ implies $g > h$", where $g = (m-1)/m^{\theta}$ and $h = m^{\theta}$ (p. 20). That needs $m - 1 > m^{2\theta}$, which holds for every $\theta < \frac{1}{3}$

once $m \geq 4$, and for $\theta = \frac{1}{4}$ once $m \geq 3$. Under Appendix A's encoding every proof has Gödel number above $32 \cdot 64^2$, so the gap is harmless; Theorems 3 and 5, which are corollaries of Theorem 1, inherit both the gap and its repair.

## 7 The Faith Is Not Just Missing

One might think Willard's systems are self-justifying merely because nobody has managed to prove step (2) in them. In fact they cannot prove it, and the reason is Oberhoff's Theorem 6 run backwards.

**Proposition K** (ours)**.** *Let $(\mathcal{F}, D)$ be a configuration with $D$ $\mathcal{F}$-sound and $\mathcal{F}$-complete, and $\mathrm{Prf}_{\mathcal{F}}^{D}$ a bounded formula that is correct in the standard model (it holds of $(h, x)$ exactly when $x$ codes a $D$-proof from $\mathcal{F}$ of the sentence coded $h$), with* SUBST *likewise correct. Let $\mathrm{Pair}(x, y)$ be a bounded formula true exactly when $x$ codes a $\Pi_1$-class sentence $\Phi$ and $y$ codes a fixed sentence logically equivalent to $\neg\Phi$; for $\Phi = \ddot{\mathsf{G}}$ that sentence is $\mathsf{N}$, with $c' = \ulcorner \mathsf{N} \urcorner$. Suppose:*

- *(H1) $\mathcal{F}$ is Level(0−) consistent: there is no $D$-proof of $0 = 1$ from $\mathcal{F}$;*
- *(H2) $(\mathcal{F}, D)$ watches;*
- *(H3) the hypotheses of Fact B, (U) included, hold, so (FP) holds;*
- *(H4) $\mathcal{F}$ proves its own Level(1) consistency for $\mathrm{Prf}_{\mathcal{F}}^{D}$ and this $\mathrm{Pair}$: $\mathcal{F} \vdash_D \forall x \forall y \forall p \forall q \neg[\mathrm{Pair}(x, y) \wedge \mathrm{Prf}_{\mathcal{F}}^{D}(x, p) \wedge \mathrm{Prf}_{\mathcal{F}}^{D}(y, q)]$.*

*Then:*

- *(a) $\mathcal{F} \vdash_D (2) \leftrightarrow \ddot{\mathsf{G}}$ and $\mathcal{F} \vdash_D [\mathsf{N} \supset \mathrm{Prov}(\bar{c}\,')] \leftrightarrow \ddot{\mathsf{G}}$;*
- *(b) $\mathcal{F} \nvdash_D \ddot{\mathsf{G}}$;*
- *(c) hence $\mathcal{F}$ proves neither step (2) nor the instance of knowing that it watches at $\mathsf{N}$, and both are true.*

*Part (b) uses (H1), (H2), the hypotheses of (FP⇒), the correctness of $\mathrm{Prf}_{\mathcal{F}}^{D}$, and the soundness and completeness of $D$. It does not use (U) or (H4).*

*Proof.* (a) By (H3), $\mathcal{F}$ proves the true bounded sentence $\mathrm{Pair}(\bar{c}, \bar{c}\,')$. With (H4), Fact A gives step (3): $\mathcal{F} \vdash_D \neg[\mathrm{Prov}(\bar{c}) \wedge \mathrm{Prov}(\bar{c}\,')]$. Now take the two directions. Step (3) and (FP) together imply $(2) \supset \ddot{\mathsf{G}}$ in every model: from (2) and (3), $\neg\mathrm{Prov}(\bar{c})$, and then (FP) gives $\ddot{\mathsf{G}}$. (FP) alone implies $\ddot{\mathsf{G}} \supset (2)$ in every model: $\ddot{\mathsf{G}}$ gives $\neg\mathrm{Prov}(\bar{c})$, which makes the antecedent of (2) false. Since $\mathcal{F}$ proves (3) and (FP), Fact A, applied with these two implications as its conclusion $\chi$, gives $\mathcal{F} \vdash_D (2) \leftrightarrow \ddot{\mathsf{G}}$. The second equivalence follows by Lemma E(b).

(b) This is Oberhoff's §4, case 1:

⇒ Suppose $\mathcal{F} \vdash_D \ddot{\mathsf{G}}$, by a proof $m$.

⇒ Since $\mathrm{Prf}_{\mathcal{F}}^{D}$ is correct, $\mathrm{Prov}(\bar{c})$ is a true $\Sigma_1$-class sentence (with witness $m$), so $\mathcal{F} \vdash_D \mathrm{Prov}(\bar{c})$ by (H2): $\mathcal{F}$ watches $\mathsf{P}(\mathsf{P})$ halt.

⇒ By (FP⇒) and Fact A, $\mathcal{F} \vdash_D \neg\ddot{\mathsf{G}}$. Then $\mathcal{F}$ proves both $\ddot{\mathsf{G}}$ and $\neg\ddot{\mathsf{G}}$, and Fact A gives $\mathcal{F} \vdash_D 0 = 1$, against (H1). ↯

(c) By (a) and (b). Truth: since no proof of $\ddot{\mathsf{G}}$ exists and $\mathrm{Prf}^D_{\mathcal{F}}$ and SUBST are correct, $\ddot{\mathsf{G}}$ is true in the standard model, step (2) is true because its antecedent is false, and the instance is true because $\mathsf{N}$ is false. □

In a system that meets these hypotheses, then, the faith step, aimed at the sentence its consistency statement pairs with $\ddot{\mathsf{G}}$, *is* the Gödel sentence, up to provable equivalence, and is true and unprovable for the same reason. Under reading (i) of Pair, $\mathsf{N}$ is a $\Sigma_1$-class sentence saying that $\mathsf{P}(\mathsf{P})$ halts, so the failure is at a $\Sigma_1$-class sentence. Under reading (ii) it is at $\neg\ddot{\mathsf{G}}$.

The proposition does not cover other formalizations of the halting sentence. Take $\mathsf{H}$ when it differs from $\mathsf{N}$. The instance of knowing that one watches at $\mathsf{H}$ is $\mathcal{F}$-provably equivalent to the transcript $\mathrm{Prov}(\bar{c}) \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$, by (FP) together with $\neg\ddot{\mathsf{G}} \leftrightarrow \mathsf{H}$. $\mathcal{F}$ proves that $\ddot{\mathsf{G}}$ implies both, as in (a). The converse would need $\mathcal{F} \vdash_D \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner) \supset \mathrm{Prov}(\bar{c}\,')$, a formalized conversion of proofs of $\mathsf{H}$ into proofs of $\mathsf{N}$. Systems with the derivability conditions prove it. For Willard's systems the question is syntactic. A tableau proof of $\mathsf{H}$ has root $\neg\mathsf{H}$ in Prenex$^*$ form, and one of $\mathsf{N}$ has root $\neg\mathsf{N}$ in Prenex$^*$ form. On the natural conversions these differ by the bounded duality between $\neg\exists h \leq x\,(A \wedge B)$ and $\forall h \leq x\,(A \supset \neg B)$, and possibly by a double negation, and Willard's p. 6 does not fix the conversion. A rewriting of closed tableaux across that difference, whose output's Gödel number is bounded by a U-Grounding term in the input's, would make the conversion a $\Pi^*_1$ sentence. If PA+ proved it, Group-2 would pass it on. Neither Willard nor we supply such a rewriting, so the question is open. Willard's own Theorem 6 meets a sharper form of the same distinction (§8).

**Corollary L** (ours)**.** *Let $A$ prove every $\Pi^*_1$ theorem of* PA+*, let every $\Pi^*_1$ theorem of $A$ be true, and let $D$ be semantic tableaux or* Tab$-U^*_1-$List*. Take $\mathrm{Prf}^D_{\mathrm{IS}_D(A)}$ to be Willard's $\mathit{SubstPrf}^D_{\mathrm{UNION}(A)}(\bar{n}_3, \cdot, \cdot)$,* Pair *to be Willard's (under either reading), and $\ddot{\mathsf{G}}$ built from that proof predicate. Assume:*

- *(E) for proofs of $\ddot{\mathsf{G}}$, under the two encoding assumptions of §2;*
- *the correctness of* SUBST*, and Willard's correctness of $\mathit{SubstPrf}^D_{\mathrm{UNION}(A)}(\bar{n}_3, \cdot, \cdot)$ in the standard model, including Eq. (7)'s bound $h \leq p$.*

*Then:*

*(a)* $\mathrm{IS}_D(A)$ *does not prove* $\ddot{\mathsf{G}}$*.*

*(b) If also $(U^*)$ holds, meaning that the uniqueness sentence (U) for this $\ddot{\mathsf{G}}$ is a $\Pi^*_1$ theorem of* PA+*, then* $\mathrm{IS}_D(A)$ *proves neither step (2) nor the instance of knowing that it watches at* $\mathsf{N}$*.*

*Proof. The proof predicate.* Willard's Eq. (7) is $\Delta^*_0$ (2005, p. 10), so $\mathrm{Prf}^D_{\mathrm{IS}_D(A)}$ is bounded. Willard introduces it as the encoding of "$p$ is a $D$-proof of $t$ from $\mathrm{IS}_D(A)$" (p. 10), and its correctness is assumed above.

*Group-2 detachment.* If $A$ proves a $\Pi^*_1$ sentence $\Phi$ by a proof $p$, then Groups 0 and 1 prove the true $\Delta^*_0$ sentence $\mathrm{Prf}_A(\ulcorner\Phi\urcorner, \bar{p})$, and with $\Phi$'s Group-2 axiom $\forall p\,\{\mathrm{Prf}_A(\ulcorner\Phi\urcorner, p) \rightarrow \Phi\}$ (Eq. 5), Fact A gives $\mathrm{IS}_D(A) \vdash_D \Phi$.

*The hypotheses.* (H1): Theorem H(a) rules out proofs of a $\Pi^*_1$ sentence and its $\Sigma^*_1$ negation, and a $D$-proof of $0 = 1$ would, by Fact A, yield proofs of every sentence, including such a pair. This holds under either reading of Pair. (H2) is Lemma I. (H3): Groups 0 and 1 prove the true bounded sentences, $\mathrm{SUBST}(\bar{n}, \bar{c})$ among them. Trichotomy is a $\Pi_1$ theorem of Peano Arithmetic, and so a $\Pi^*_1$ theorem of PA+, which detachment hands to $\mathrm{IS}_D(A)$. (E) is assumed. Under $(U^*)$, (U) arrives by the same detachment. (H4) is literally the Group-3 axiom, Eq. (8), once $\mathrm{Prf}^D_{\mathrm{IS}_D(A)}$ and Pair are Willard's. Part (a) is Proposition K(b), which does not use (U). Part (b) is Proposition K(c). □

($U^*$) is a transposition and is ours: Willard asserts (U) for $\Delta_0^-$ substitution and PAX (2001, p. 39), and we use it for the $\Delta_0^*$ substitution formula of 2005 (p. 32) and PA+. The two bulleted assumptions are open hypotheses of both parts: (E) rests on our reading of Willard's encoding, and the correctness of *SubstPrf* is Willard's assertion. Both are used in the metatheory only. $\mathrm{IS}_D(A)$ is never asked to prove that its Group-3 formula is its own proof predicate.

**Corollary M** (ours)**.** *Let $A \supseteq$ PAX be regularly consistent, let $D$ be semantic tableaux, and let $\ddot{\mathsf{G}}$ be built from the proof predicate of* IS($A$)*'s Group-3 axiom. Here* $\mathsf{N}$ *is* $\neg\ddot{\mathsf{G}}$ *and* $c' = \ulcorner\neg\ddot{\mathsf{G}}\urcorner$*. Assume (E) for proofs of* $\ddot{\mathsf{G}}$ *and the correctness of* SUBST *and of that proof predicate, as in Corollary L. Then* IS($A$) *does not prove both step (2) and the Level(0−) form of step (3),* $\forall p\,\forall q\,[\mathrm{Prf}(\bar{c}, p) \wedge \mathrm{Prf}(\bar{c}\,', q) \supset \exists r\,\mathrm{Prf}(\ulcorner 0 = 1\urcorner, r)]$.

*Proof.* If IS($A$) proved both, then with its Group-3 axiom $\forall r\,\neg\mathrm{Prf}(\ulcorner 0 = 1\urcorner, r)$ it would prove $\neg[\mathrm{Prov}(\bar{c}) \wedge \mathrm{Prov}(\bar{c}\,')]$, which is step (3) at the one pair $(\ddot{\mathsf{G}}, \neg\ddot{\mathsf{G}})$. That is all the proof of Proposition K(a) uses of (H4). The other hypotheses hold in the setting of Willard's own Theorem 7.2. (H1) is Theorem H(b). (H2): IS($A$) proves every true $\Delta_0^-$ sentence (2001, p. 39) and names every number by a constant, so for a true $\Sigma_1^-$ sentence it proves the true $\Delta_0^-$ instance at the witnesses, and Fact A adds the existential quantifiers. (H3): trichotomy and (U) are $\Pi_1^-$ theorems of PAX (2001, p. 39, for (U)), inherited by the same detachment, and (E) is assumed. The uniqueness sentence (U) needs no transposition here. Then the proof of Proposition K(a) gives $\mathrm{IS}(A) \vdash (2) \leftrightarrow \ddot{\mathsf{G}}$, hence $\mathrm{IS}(A) \vdash \ddot{\mathsf{G}}$, against Proposition K(b). □

So for the 2001 system the faith fails in at least one of two places, and we do not know which.

The proposition is a contrapositive and claims no novelty of method. Its sibling in Willard is 2001 Theorem 7.2 (p. 38, `full`): no consistent $\alpha$ with a $\Delta_0^-$ proof predicate, an $\alpha$-sound and $\alpha$-complete $D$, and all $\Pi_1^-$ theorems of PAX can prove the canonical reflection principle $\forall y\,\{\mathrm{Prf}_\alpha^D(\ulcorner\Upsilon\urcorner, y) \supset \Upsilon\}$ for every $\Pi_1^-$ sentence $\Upsilon$. Its proof uses the same fixed point and the same Fact A in place of the derivability conditions; "The above result is different from Löb's Theorem ... chiefly because we do not assume that $\alpha$ recognizes either Addition or Multiplication as total functions" (p. 38, text layer). Willard's 1994 framing already sets out the alternatives: "every axiom system either fails to prove its consistency or fails to meet the full Hilbert-Bernays conditions" (TR 93-10, p. 6, verbatim). What the computational reading adds is *which* condition fails, at *which* sentence, and that the failing instance is the Gödel sentence in disguise. What Willard adds, and no contrapositive can supply, is that systems on the far side of the alternative *exist* (Theorem H).

## 8 The Faith, Bought: Willard's Theorem 6

Willard's 2005 paper ends by doing the experiment. Take the transcript statement of §4, bound it, and see what it costs to switch it on.

Let $\mathrm{Log}^k$ be the $k$-fold base-2 logarithm, rounded down, and $\Upsilon(k, m)$ the true $\Sigma_1^*$ sentence $\exists z\ \mathrm{Log}^k(z) \geq \bar{m}$, which says that a number at least $2_k^m$ exists, where $2_k^m$, the least $z$ with $\mathrm{Log}^k(z) \geq m$, is roughly a tower of $k$ exponentials topped by $m$ (2005, Definition 2, p. 11). $\mathrm{NS}_D^{k,m}(A)$ is $\mathrm{IS}_D(A)$ with $\Upsilon(k, m)$ added to Group-0 and Group-3's "this" adjusted to the larger system (Definition 3, p. 11). Both systems prove $\Upsilon(k, m)$; only one of them has it as an axiom. Let $\mho(s, b)$ be the bounded Gödel sentence "no proof $p$ of me with $\mathrm{Log}^s(p) < b$" (Eq. 24, p. 33), and *CheckProof*$(t, p, q)$ the $\Delta_0^*$ formula "$p$ proves $t$, and $q$ proves $\exists p\,\mathrm{Prf}(t, p)$" (Definition 6, p. 33). Lemma 4 (p. 33, `full`, resting on Lemma 3, a proof `sketch` that checking a proof is quasi-polynomially cheap) says that for $k$ large Peano Arithmetic proves the $\Pi_1^*$ sentence (Willard writes Peano Arithmetic; the sentence is in

the U-Grounding language, so the theory meant is PA+)

$$\forall p\,\forall n\,\exists q < n\,\{[\mathrm{Log}^{k-1}p < m \wedge \mathrm{Log}^{k}n \geq m \wedge \mathrm{Prf}(\ulcorner \mho(k-1,m) \urcorner, p)] \\ \rightarrow \mathit{CheckProof}(\ulcorner \mho(k-1,m) \urcorner, p, q)\}. \quad (27)$$

This is the literal transcript of step (2), bounded. For every short proof $p$ of the Gödel sentence and every number $n$ that is big enough, a transcript $q$ exists below $n$, and $q$ proves the $\Sigma_1^*$ sentence (25), $\exists p\,\mathrm{Prf}(\ulcorner \mho(k-1,m) \urcorner, p)$: "$\mho$ is provable". Unlike the halting sentence for $\ddot{\mathsf{G}}$, this is not equivalent to the negation of the Gödel sentence. $\mho$ speaks only of short proofs, so $\neg\mho$ says that a short proof exists, while (25) says that some proof exists. Because the transcript is bounded by $n$, the statement is $\Pi_1^*$, and the system inherits it from PA+ through Group-2. What is missing is $n$. (Willard states Lemma 4 for the proof predicate of $\mathrm{NS}_D^{k,m}(\mathrm{PA}+)$. Nothing in the sketch of Lemma 3 is specific to the added axiom, so we expect the same bounded faith step for $\mathrm{IS}_D(\mathrm{PA}+)$'s own proof predicate, but Willard does not state it.)

**Theorem N** (Willard 2005, Theorem 6, p. 34, `full`; gap at step (31), below)**.** *For* $D = \mathrm{Tab}{-}U_1^*{-}\mathrm{List}$ *there is* $k_0$ *such that for every* $k > k_0$, $\mathrm{NS}_D^{k,m}(\mathrm{PA}+)$ *is inconsistent.*

Willard's proof is a seven-step tableau list $W$ (pp. 34–36), and, up to the gap at its fourth step discussed below, it is Oberhoff's Theorem 6 with the faith supplied:

- $\Rightarrow$ (27), inherited from Peano Arithmetic: the faith step, below $n$.
- $\Rightarrow$ (28) $\mathrm{Pair}(\ulcorner \mho \urcorner, \ulcorner \neg \mho \urcorner)$, a true bounded sentence.
- $\Rightarrow$ (29) the Group-3 axiom, and (30) its instance: step (3).
- $\Rightarrow$ (31), a step Willard calls "mostly notational": no $p$, $q$ satisfy $\mathit{CheckProof}(\ulcorner \mho \urcorner, p, q)$. Then (32), chaining with (27): no short proof of $\mho$ coexists with a big $n$.
- $\Rightarrow$ (33) $\Upsilon(k,m)$, an axiom, supplies $n$: no short proof of $\mho$ exists. This is $\mho(k-1,m)$ itself, which the seventh step derives.
- $\Rightarrow$ Because $\Upsilon(k,m)$ cost nothing, $W$ is short: for $k > k_0$ its Gödel number is below $2_{k-1}^m$, so $W$ is itself one of the short proofs that $\mho(k-1,m)$ says do not exist. The system watches $W$ and refutes $\mho$. $\lightning$

Willard's fourth step meets a sharper form of the distinction drawn after Proposition K. He derives (31) from (30) on the ground that "the definitions of CheckProof and $\mho(k-1,m)$ imply that CheckProof$(\ulcorner \mho \urcorner, p, q)$ is logically equivalent to Equation (30)'s curly bracket expression" (p. 35, text layer), that is, to $\mathrm{Prf}(\ulcorner \mho \urcorner, p) \wedge \mathrm{Prf}(\ulcorner \neg \mho \urcorner, q)$. By Definition 6, as printed (we checked the page image), $q$ proves (25), not the sentence Pair pairs with $\mho$, under either reading. The two sentences have different codes and differ in content too: (25) has neither the substitution nor the bound $\mathrm{Log}^{k-1}(p) < m$, and a long proof of $\mho$ makes (25) true and $\neg\mho$ false. So step (31) is a gap at our granularity, and we found no reading of Willard's text under which it disappears. It could be closed in either of two ways:

- redefine *CheckProof* so that $q$ proves the sentence Pair pairs with $\mho(k-1,m)$, and supply for such $q$ a length bound to replace Lemma 3, from which Corollary 1 and Lemma 4 would then follow;
- keep Definition 6 and prove, within the bound $n$, that a proof of (25), given the short $p$, converts into a proof of that sentence.

Lemma 3's sketch gives neither. It produces a short proof that $p$ is a proof, which supports the printed *CheckProof*. We conjecture that the first repair works, since a proof of $\neg\mho$ can exhibit $p$, check the bounded facts $\mathrm{SUBST}(\bar{n}, \ulcorner\mho\urcorner)$, $\mathrm{Log}^{k-1}(p) < m$ and $\mathrm{Prf}(\ulcorner\mho\urcorner, p)$, and generalize. No length estimate for it is proved, by Willard or here. We therefore list Theorem 6 as `full` with a gap at step (31).

Willard calls the length estimate "quite routine" (p. 36). He measures the difference between the two systems by a constant: proof lengths in $\mathrm{NS}_D^{k,m}(A)$ and $\mathrm{IS}_D(A)$ differ by at most some $C > 2_k^m$, "an enormous quantity" (p. 31). Why the argument does not transfer to $\mathrm{IS}_D(A)$ can be seen with Willard's own engine (our observation). Under the hypothesis of Theorem H(a) with $A = \mathrm{PA}+$, all axioms of $\mathrm{IS}_D(\mathrm{PA}+)$ are true $\Pi_1^*$ sentences: Group-3 by Theorems 3 and 5. So the system is $\mathrm{Normed}(a, 1)$ for every $a > 1$. $\mathrm{IS}_D(\mathrm{PA}+)$ must prove $\Upsilon(k, m)$ as a lemma. Let $p > 1$ be a proof of it, and take $a = p$. Then $p \leq \lceil a/1 \rceil + 1$, and $\theta$-compactification with $\theta = \frac{1}{4}$ (Theorems 2 and 4) makes $\Upsilon(k, m)$ $\mathrm{App}\exists(p^{1/4})$-valid: it has a witness at most $p^{1/4}$. The least witness is $2_k^m$, so $p \geq (2_k^m)^4$. In a tableau list, the components up to the lemma form a list proof of it, and they are a prefix of the whole list's byte string, so the whole list has a Gödel number at least $p$. That lies far beyond the bound $2_{k-1}^m$ of which $\mho(k-1, m)$ speaks. "A Kleene-like Group-3 axiom has a different connotation when it refers to a short proof, referencing itself, than when it refers to a long such proof" (p. 32, text layer). The asymmetry is exact: adding a true $\Pi_1^*$ sentence instead of the $\Sigma_1^*$ sentence $\Upsilon(k, m)$ preserves consistency (p. 37, stated as a generalization of Theorem 5). A universal axiom constrains; an existential axiom supplies a number, and the number is what the faith step was missing. Remark 8 (p. 36, `stated-only`) extends the theorem to semantic tableaux and other cut-free methods, to empty $A$, and to systems that drop the two Group-zero axioms declaring addition and doubling total.

## 9 The Price of Faith

If withholding faith is what lets $\ddot{\mathrm{W}}$ survive, the second incompleteness theorem should return as soon as the system can afford the transcripts. Willard spent the second half of his career proving that it does. The price depends on the language and on how much the proof system may reuse.

### Squaring

Give a tableau system multiplication as a total function. Then a proof can introduce $u_0 = 2$, $u_{i+1} = u_i \cdot u_i$, and so reach $u_n = 2^{2^n}$, in $O(n)$ steps. The 2001 system with multiplication added, $\mathrm{ISMULT}(A)$, "satisfies an analogue of" Assertion ++ but has no analogue of Lemma 4.2, for exactly this reason (2001, Remark 4.5, p. 21). On the negative side Willard located the failure precisely (2000, §5, quoted in the codified statement §5.3):

> Many readers will smile with amusement when they learn the true reason that the Semantic Tableaux version of the Second Incompleteness Theorem breaks down when Multiplication is changed from a total function into a 3-way relation. It is essentially that Lemma 8 and its short proof then become no longer valid. ... Without it, our proof of the Semantic Tableaux version of the Second Incompleteness Theorem collapses entirely!

Lemma 8 (2000, p. 11, `full`) is the squaring construction: a tableau proof from $Q + V$, of length $O([\log\log z]^C)$, that a number about as large as $z$ exists. With it, the second theorem comes back, at Level(0−):

- no consistent *finite* extension of $Q + V$, for an explicit $\Pi_1$ sentence $V$ with $Q + V$ consistent, proves its own semantic tableaux consistency (2002a, Theorems 3.4–3.5, p. 11, `full`);

- every consistent extension of $T$-1 or $T$-2, the textbook and Wilkie–Paris axiomatizations of $I\Sigma_0$, is unable to prove its own semantic tableaux consistency (2007, Theorem 6, p. 29, `full`);
- any consistent extension of $I\Sigma_0$ with the Conventional Deciphering Property (a finite subset proves, of each axiom's canonical numeral, that it is an axiom: 2002a, Definition 5.1, p. 20) cannot prove that there is no tableau proof of $0 = 1$ from itself (2002a, Theorem 6.4, p. 25, `sketch`).

In 1981 Paris and Wilkie asked whether $I\Sigma_0$ satisfies the second incompleteness theorem for cut-free deduction. 2002a presented Theorem 6.4 as an answer; 2007 records that the 2002 paper covered only the textbook induction scheme ("which we did not examine in our year-2002 article", of the Wilkie–Paris scheme: 2007, abstract, text layer), and its Theorem 6 covers both. On the positive side, Definition 5 becomes "impossible to satisfy" once multiplication is admitted as a ninth function symbol (2005, Remark 4, p. 21, `stated-only` as to the reason: "intuitively because multiplication has a faster growth property than addition").[1]

## Subroutines

A Hilbert-style proof can prove a lemma once and use it many times. Modus ponens is a subroutine call. A tableau cannot do this; every use must be inlined. Inlining can be astronomically expensive: by Gentzen, the tableau proof exists, but "$t$'s length can certainly be super-exponentially longer" than the proofs it combines (2002a, Theorem 2.2, p. 7, `sketch`), and Boolos gave a famous small example of the blowup [Boolos 1984]. With subroutines, a little growth is amplified into a lot. Willard saw this in 1994 (TR 93-10, p. 7, verbatim):

> These metaphors lose meaning when cuts are present because Addition can then be employed to *implicitly define* a sequence that grows actually faster than $a_0, a_1, a_2, a_3 \ldots$. Solovay's proof that IS(A) is incompatible with deductive cuts [So94] rests on this fast growth property. (Actually, IS(A) can support a cut rule for $\Delta_0$ sentences, and Solovay's theorem relies on the cut rule for sentences with unbounded quantifiers.)

The boundary depends on the *complexity* of the lemmas, and Willard mapped it:

- a cut rule for bounded ($\Delta_0$) sentences: compatible with IS($A$), as the parenthesis above says (TR 93-10, p. 7; the codification records the construction in a footnote on p. 35, obligation O56);
- lemmas with one unbounded quantifier block ($\Pi_1^*$ or $\Sigma_1^*$), addition total: self-justifying at Level(1) (2005, Theorem 5, `full`);
- lemmas with two blocks ($\Pi_2^*$, respectively $\Sigma_2^*$), addition total: for explicit $\Pi_1^*$ theorems $V_A$, $V_B$ of Peano Arithmetic, no consistent system in the U-Grounding language containing $V_A$ (resp. $V_B$) proves its own Level(0−) consistency under Tab−$\Pi_2^*$ (resp. Tab−$\Sigma_2^*$) lists (2004, Theorem 2, p. 344, `sketch`; stated for finite systems "for the sake of simplifying our proof");
- unrestricted lemmas (Hilbert deduction), successor total together with the $\Pi_1$ algebra of $+$ and $\times$: Solovay's theorem below.

[1] What multiplication destroys is the flat consistency axiom and canonical reflection, not every self-referential construction. Willard's ISM($A$) recognizes multiplication as total and is consistent whenever $A$ is regularly consistent, "by a routine generalization" of his proof, but its self-referential axiom is weaker, and it cannot prove the flat statement that no tableau proof of $0 = 1$ exists (2001, §6, p. 36; codified statement §7.2).

The unrestricted case runs through the classical shortening of definable cuts. In outline (a textbook gloss, not Willard's text): with lemmas, one defines a set of numbers and proves once that it is closed under successor; then, by a short proof reusing that lemma, that a smaller set is closed under addition, and then that a smaller set still is closed under multiplication. On the last set the system can afford transcripts, and Gödel's argument runs there. Pudlák showed that a consistent extension of Robinson's $Q$ proves neither its Hilbert consistency nor its consistency restricted to any definable cut [Pudlák 1985] (Willard 2006a, Theorem 1, p. 4, `cited`). Solovay's theorem, communicated to Willard by telephone in April 1994 and never published by Solovay, states the conclusion (2006a, Theorem 2, p. 4, `cited`): a consistent system that formalizes arithmetic with three-place predicates for $+$ and $\times$, proves successor total, and proves the $\Pi_1$ associative, commutative, distributive and identity laws for those predicates cannot prove that there is no Hilbert proof of $0 = 1$ from itself. With unrestricted subroutines, the price of faith falls to successor together with the ring laws for $+$ and $\times$ as relations: neither addition nor multiplication need be total.

So the Hilbert-line self-justifying systems give up successor itself. They name numbers with constants $C_1, C_2, \ldots$, and the naming convention *is* the system (codified statement §5.5). Incremental naming $C_i = C_{i-1} + 1$ is how 2006a reads 2001's ISREF($A$), which is consistent when $A$ is regularly consistent (2001, Theorem 3.4, p. 13, `full`) and whose Group-3 is the tangibility schema (2001, Eq. 10), not a flat consistency axiom: "The term 'incremental naming convention' had not appeared in our article [46]. However, its Theorem 3.4 had implicitly presumed the presence of an incremental naming method" (2006a, p. 22, text layer). Additive naming $C_i = C_{i-1} + C_{i-1}$ works and keeps reaching new numbers (ISCE($A$), for consistent $A$ with true $\Pi_1^-$ theorems and $\Delta_0^-$ Hilbert proof predicate: 2006a, Theorem 3, p. 6, `full`); multiplicative naming $C_i = C_{i-1} \cdot C_{i-1}$ fails for any system that contains the convention's axioms, proves Peano Arithmetic's $\Pi_1^-$ theorems and has Willard's Concise Encoding property (2006a, Theorem 4, p. 7, `full`). Remark 3, quoted in §4, explains why.

### Other ways to buy faith

Excluded middle as a logical axiom, $\Phi \vee \neg\Phi$ at any node for any $\Phi$ (Xtab, 2020, p. 10), is what TR 93-10 had already defined as "a tableaux proof with cuts" (printed p. 33, as recorded in codified statement §7.3). Then $\mathrm{IS}_{\mathrm{Xtab}}(\beta)$ is automatically inconsistent whenever $\beta$ proves the usual $\Pi_1^*$ associative, commutative, distributive and identity laws for addition and multiplication (2020, Theorem 4.5, p. 12, `sketch`). And letting $\ddot{\mathsf{W}}$ say more about itself brings the second theorem back: there is a $\Pi_1^*$ theorem $W$ of Peano Arithmetic such that no consistent finite system in the U-Grounding language containing $W$ can prove its own Level($2^+$) tableaux consistency, the absence of proofs of a Boolean combination of $\Pi_2^*$ and $\Sigma_2^*$ sentences together with its negation (2004, Theorem 1, p. 342, `full`, although the codification records that one step is deferred, in a footnote saying it "should not be considered a formal proof": codified statement §7.5).

| Proof system | Growth | Consistency asserted | Outcome | Source, status |
|---|---|---|---|---|
| tableaux, or lists of $\Pi_1^*/\Sigma_1^*$ lemmas | + total, × a relation | Level(1) | self-justifying | 2005 Thms 3, 5, `full` |
| same | same, plus one axiom $\exists z\,\mathrm{Log}^k z \geq m$ | Level(1) | inconsistent | 2005 Thm 6, `full` (Lemma 3 `sketch`; gap at step (31), §8) |
| tableaux | × total | Level(0−) | G2 | 2002a Thm 3.5 `full`; 6.4 `sketch`; 2007 Thm 6 `full` |
| lists of $\Pi_2^*$ or $\Sigma_2^*$ lemmas | + total | Level(0−) | G2 | 2004 Thm 2, `sketch` |
| tableaux | + total | Level($2^+$) | G2 | 2004 Thm 1, `full` (one deferred step) |
| tableaux + excluded middle | + total, × laws | Level(1) | inconsistent | 2020 Thm 4.5, `sketch` |
| Hilbert | successor total, $\Pi_1$ laws of +, × | Level(0−) | G2 | Solovay, `cited` |
| Hilbert | additive names only | Level(0−) | self-justifying | 2006a Thm 3, `full` |
| Hilbert | multiplicative names | Level(0−) | G2 | 2006a Thm 4, `full` |

Read the table as Willard read it. Negative rows assert the weakest consistency notion and positive rows the strongest, by the rule of §2, so each row is as strong as its side allows. "Self-justifying" in a positive row means that the system proves its own consistency and is consistent under the hypothesis on its base theory (Theorem H; 2006a, Theorem 3). The affirmative results are mostly `full`; the negative side contains sketches and one unpublished theorem. The boundary is sharply located in some coordinates and only bracketed in others (codified statement §8.2).

## 10 Trust Within Reach

A system *trusts* its proofs if it proves canonical reflection, $\forall y\,\{\mathrm{Prf}_\alpha(\ulcorner\Psi\urcorner, y) \supset \Psi\}$, for every $\Pi_1$-class $\Psi$. That is too much to ask: 2001 Theorem 7.2 (§7) forbids it for any consistent system with a bounded proof predicate that inherits the $\Pi_1^-$ theorems of PAX. What Willard's systems can prove is *tangibility reflection* (2001, Definition 1.2, p. 4, Eq. 2):

$$\forall x\ \{\ [\exists y\,\mathrm{Prf}_\alpha(\ulcorner\Psi\urcorner, y) \wedge \mathit{Tangible}(x)] \supset \Psi^x\ \},$$

where $\Psi^x$ is $\Psi$ with its unbounded universal quantifiers bounded by $x$, and *Tangible* is one of $\mathit{TangPred}(x) \equiv \exists v\, x < v - 1$, $\mathit{TangDiv}_k(x) \equiv \exists v\, x < v/k$, or $\mathit{TangRoot}_k(x) \equiv \exists v\, x < v^{1/k}$. ISREF($A$), a Hilbert-style system without addition total, proves the *TangPred* form for every prenex* $\Psi$, given that $\forall y\, \mathit{Size}(y) < y - 1$ (2001, Theorem 3.5, p. 14, `full`). The tableau system $\mathrm{IS}^\lambda(A)$ has a Hilbert proof of the $\mathit{TangRoot}_k$ form when $0 < \lambda < 1$, $k > 1/(1-\lambda)$, and $A$ proves a stated $\Pi_1^-$ sentence (Theorem 6.1, p. 34, `full`), and a tableau proof by cut elimination (Corollary 6.2, p. 35, `full`); its consistency is proved for $\lambda = 3/4$ (Theorem 5.1, p. 28, `full`), where the condition reads $k > 4$. In computational terms: *I trust my verdicts on every input for which there is memory to spare.* For a $\Sigma_1$-class claim the bound costs nothing, since $\Psi \equiv \Psi^0$; taking $\Psi$ to be $0 = 1$ gives back the flat statement that no proof of $0 = 1$ exists (2001, p. 8).

## 11 Taking Off the Glasses

Oberhoff's §5 takes the glasses off $\ddot{\mathsf{G}}$ and finds "I am not provable" underneath. Take them off $\ddot{\mathsf{W}}$ and you find "nothing refutes me", that is, "I am consistent". Over extensions of $I\Sigma_1$ with Hilbert deduction, the metatheory shows that $\mathcal{F}_0 + \ddot{\mathsf{W}}$ derives a contradiction (Proposition G), so $\mathsf{C}(\mathsf{C})$ halts. In Willard's systems that refutation would have to establish the existence of a transcript-sized object from inside a proof smaller than the object. The snake cannot swallow its tail, because its mouth is smaller than its body.

So self-reference is not where the trouble lies. The fixed point builds $\ddot{\mathsf{W}}$ in Willard's systems as easily as in Peano Arithmetic. What costs something is *self-simulation*: running yourself in your head and writing down the result. The second incompleteness theorem needs both. The first needs self-reference and watching, and, in Rosser's form, one $\Pi_1$ fact about a loop. That is why Willard's systems obey it.

Oberhoff found that his classical and computational proofs were "the exact same". The same holds here. Willard's arithmetic (a least pair of contradictory proofs, values below $b \cdot 2^{\text{U-Depth}}$, five bits per symbol) and the computational picture (a shortest refutation, the memory a proof occupies, a walk down the tree) are one argument. The arithmetic of Gödel numbers is the bookkeeping of memory.

## 12 Refactoring the Incompleteness Theorems, Again

Oberhoff's Theorem 7 is two theorems. Stated as properties of a configuration $(\alpha, D)$, they are as follows. Property 7b, with its clause for syntactic negations, is our formulation; the clause is there so that a failure at $\neg\ddot{\mathsf{G}}$ counts.

> **7a** (Watching, Definition C). $\alpha \vdash_D \sigma$ *for every true* $\Sigma_1$*-class sentence* $\sigma$*; in particular, for the halting sentence of each halting computation.*
>
> **7b** (Knowing that one watches, Definition D). $\alpha \vdash_D \sigma \supset \mathrm{Prov}_\alpha^D(\ulcorner\sigma\urcorner)$*, as a schema, for every* $\Sigma_1$*-class sentence* $\sigma$ *and for the syntactic negation* $\sigma$ *of every* $\Pi_1$*-class sentence.*

Property 7a holds for every configuration in this essay: for extensions of Robinson's $Q$ with Hilbert deduction (standard), for $\mathrm{IS}_D(A)$ (Lemma I), and for $\mathrm{IS}(A)$ with $A \supseteq \mathrm{PAX}$ (§6). Property 7b holds for recursively axiomatized extensions of $I\Sigma_1$ with their canonical provability predicate [Hájek–Pudlák 1993], where the derivability conditions also carry each written form of a $\Sigma_1$ fact to the other. Property 7a gives Oberhoff's Theorem 4 and the $\ddot{\mathsf{G}}$ half of his §4. Rosser's form also needs the $\Pi_1$ loop sentence of footnote 12, which Willard's systems get from their base theory (§6, for $A \supseteq \mathrm{PA}+$). Property 7b is what Theorem 6 takes on faith, and it does not follow from 7a:

- $\mathrm{IS}_D(A)$, under the hypotheses of Corollary L and its transposed assumption ($U^*$), satisfies 7a, proves its own Level(1) consistency, and fails 7b at one sentence, $\mathsf{N}$, where the failing instance is provably equivalent to the Gödel sentence (Proposition K). Under reading (i) of Pair, $\mathsf{N}$ is a $\Sigma_1$-class sentence saying that $\mathsf{P}(\mathsf{P})$ halts. Under reading (ii) it is $\neg\ddot{\mathsf{G}}$.
- $\mathrm{IS}(A)$, under the hypotheses of Corollary M, fails either step (2), and with it 7b at $\neg\ddot{\mathsf{G}}$, or the Level(0−) form of step (3). We do not know which.
- Whether either system fails 7b at a different formalization $\mathsf{H}$ of the halting sentence is open (the remark after Proposition K).

By Lemma E(a) these systems satisfy 7b at every *true* sentence of either kind, so the failure is confined to runs that do not happen. Willard's Theorem 6 (§8) shows a bounded transcript form of 7b, at "$\mho$ is provable", to be a $\Pi_1^*$ theorem of PA+, inherited through Group-2. Whether it can be made to reach the sentence Pair pairs with $\mho$ is the gap noted there. What the system lacks is one number large enough to switch it on, and granting that number as an axiom makes the system inconsistent.

So Gödel's work splits into three parts, not two. The first is programming the integers (7a). The second is teaching the integers that they have been programmed (7b). The third is the victory laps. Willard's self-justifying systems live in the gap between the first and the second. The second incompleteness theorem is a victory lap only for a runner who has already paid for 7b. Willard's accounting does not give one price. It gives four boundaries (§9):

- tableaux without lemmas, multiplication total: the second theorem holds at Level(0−) for consistent finite extensions of $Q + V$ and for consistent extensions of $I\Sigma_0$'s two axiomatizations (2002a, Theorem 3.5; 2007, Theorem 6; `full`);
- lemmas with one unbounded quantifier block, addition total and multiplication a relation: $\mathrm{IS}_D(A)$ is self-justifying at Level(1), and consistent whenever $A$'s $\Pi_1^*$ theorems are true (2005, Theorem 5, `full`);
- lemmas with two blocks, addition total: for explicit $\Pi_1^*$ theorems $V_A$, $V_B$ of Peano Arithmetic, no consistent finite system containing $V_A$ (respectively $V_B$) proves its own Level(0−) consistency under $\Pi_2^*$ (respectively $\Sigma_2^*$) lemma lists (2004, Theorem 2, `sketch`; finiteness "for the sake of simplifying our proof");
- unrestricted lemmas: successor total together with the $\Pi_1$ ring laws for $+$ and $\times$ as relations already suffices (Solovay, `cited`).

At the finest grain, one large number granted as an axiom is enough (2005, Theorem 6).

Moral of the story: you may declare yourself a very stable genius, and Willard's theorems will vouch for you provided your teachers told you only truths, you cannot multiply, and you reuse only lemmas with one unbounded quantifier block. Learn to multiply, and (for the theories Willard treats) even proofs without lemmas are too much. Reuse lemmas with two blocks, and (by a sketch of Willard's) addition is already too much. Reuse any lemma you like, and (by Solovay's unpublished theorem) successor with the ring laws for $+$ and $\times$ is too much. And you must never be told, as an axiom, that a big enough number exists.

## 13 What Self-Justification Does Not Buy

Willard was more careful about the limits of his results than most of his readers have been, and an honest retelling must carry his caveats.

- **Relative consistency.** Theorem H assumes that the base theory's $\Pi_1$-class theorems are true. Nothing here shows any system consistent outright; the systems are as trustworthy as their base. Willard: "if $A$ = Peano Arithmetic, IS(A) can prove all the $\Pi_1^-$ theorems of Peano Arithmetic, and its consistency will be as certain as that of Peano Arithmetic!" (2001, p. 11, text layer).
- **A reduced question.** Willard (2016, Remark 4.9, p. 13) splits Hilbert's question in two: can a system verify its own consistency in a robust sense, and "Can logic systems 'appreciate' their

own consistency in some REDUCED sense, that is diluted but not fully immaterial?" (text layer). The answer to the first is no, by Gödel, Hilbert–Bernays and Pudlák–Solovay. Everything above answers the second.

- **A one-line proof.** A system that proves its consistency from its own axiom $\ddot{\mathsf{W}}$ has "essentially a 1-line proof" (2011, Remark 6.16, p. 28), which is instinctive faith rather than proof-justification. The substantive justification lies outside: the theorem that *Probe* defeats every refutation, proved in the metatheory under the hypothesis on $A$.

- **Almost cheating.** Willard raised the objection against himself: "Is it not almost cheating when an axiom system verifies its own consistency by using $\oplus$'s formalized 'I am consistent' axiom as an intermediate step, to verify its own consistency? After all, such a technique can verify its own consistency only in a technically purely legalistic sense" (2016, §8). His answer, *Platonic Stability*, is that a thinking being needs no sophisticated justification of its own reasoning, only a formalism in which presuming its own consistency provably will not spin it into inconsistency. Its formal statement, Corollary 8.2 (p. 28), opens "Let us assume Conjecture 6.6 is correct (as we are almost 100 % certain it is)"; Conjecture 6.6 (p. 23) is `stated-only`.

- **The first theorem stands.** Under the hypothesis of Theorem H(a), with $A$ proving the $\Pi_1^*$ theorems of PA+, $\mathrm{IS}_D(A)$ decides neither $\ddot{\mathsf{G}}$ nor $\ddot{\mathsf{R}}$. $\ddot{\mathsf{G}}$ is unprovable by Corollary L(a), with its encoding and correctness assumptions but without $(U^*)$. $\neg\ddot{\mathsf{G}}$ is unprovable by honesty (Lemma J, under its reading of Pair), and Rosser's form holds by §6. For $\mathrm{IS}(A)$, Willard's TR 93-10 says that it satisfies "Godel's First but not Second Incompleteness Theorem" (p. 6); the essay does not prove this for its $\ddot{\mathsf{G}}$ and $\ddot{\mathsf{R}}$.

- **Interpretation versus source.** The vocabulary of watching and knowing, the distinctions among $\neg\ddot{\mathsf{G}}$, $\mathsf{H}$ and $\mathsf{N}$ and between the two readings of Pair, the program $\mathsf{C}$ as a reading of Willard's Group-3, the horizon picture, Lemmas E–J, Proposition K and its corollaries, the reading of Willard's Theorem 6 as Oberhoff's Theorem 6 with the faith supplied, and the repair of its fourth step are ours, not Willard's formulations. His own words for the mechanism are "adequate memory" (1994) and the comparison, in 2006a Remark 3, between a proof's Gödel number and the integer the proof shows to exist.

(The cake promised in Oberhoff's abstract is still on its way. We cannot prove that it exists: it is one multiplication away.)

# A Provenance

Page numbers are the printed page numbers of the author copies held in `nachlass/papers/` in the repository; for 2004 they are the volume's. The PDF page index differs for TR 93-10 (by 2), for 2011, 2016 and 2020 (by 1) and for 2004, and the codification sometimes cites the index instead. Statuses are those of `nachlass/codification/registry/results.md`. "Verbatim, image" marks quotations checked against rendered page images; "text layer" marks quotations checked only against the extracted text.

| Paper | Item | Page | Content used here | Status |
|---|---|---|---|---|
| TR 93-10 | — | 6 | philosopher's hedge; G1 but not G2; "either fails … or fails to meet"; $\mathit{Turing}(x, y, z)$ | verbatim, image |
| TR 93-10 | — | 7 | adequate memory; $a_{300}$; cuts and Solovay | verbatim, image |
| TR 93-10 | — | 33 | tableaux with $\Phi \vee \neg\Phi$ (Xtab) | via codified §7.3 |
| 2000 | Lem 8; §5 | 11; — | squaring construction; "the true reason" | full; verbatim (codified §5.3) |
| 2001 | Def 1.2 | 4 | tangibility predicates, Eq. 2 | — |
| 2001 | §2 | 9–12 | grounding functions; IS(A) groups; regular consistency | text layer |
| 2001 | Thm 3.4, 3.5 | 13–14 | ISREF($A$) consistency; *TangPred* reflection | full |
| 2001 | Lem 4.1, 4.2 | 17, 19 | constants below $\sqrt{y}$; parameters below $y$ | full; sketch |
| 2001 | Thm 4.3; Rem 4.4 | 20 | IS(A) consistency; Assertion ++ | full; verbatim, image |
| 2001 | Rem 4.5 | 21 | ISMULT($A$) has no analogue of Lem 4.2 | — |
| 2001 | Lem 4.6, 4.8 | 22–23 | affirmative branch cannot close; least refutation has one | full |
| 2001 | Case 11 | 28 | "substantially more than six bits" | image (extraction record) |
| 2001 | Thm 5.1, 6.1; Cor 6.2 | 28, 34–35 | $\mathrm{IS}^{\lambda}(A)$ at $\lambda = 3/4$; $\mathit{TangRoot}_k$ reflection | full |
| 2001 | §6 | 36 | ISM($A$) | routine generalization (Willard) |
| 2001 | Lem 7.1; Thm 7.2, 7.3 | 37–40 | closure; no canonical reflection, fixed point (63), (67), (U); $\mathrm{XIS}^{\lambda}$ | full; text layer |
| 2001 | App B | 51–52 | byte encoding of Hilbert and tableau proofs; (E) | text layer |
| 2002a | Thm 2.2 | 7 | cut elimination, super-exponential length | sketch |
| 2002a | Thm 3.4, 3.5 | 11 | $Q + V$: G2 for finite extensions, tableaux | full |
| 2002a | Def 5.1; Thm 6.4 | 20, 25 | Conventional Deciphering; $I\Sigma_0$ | —; sketch |
| 2002c | — | 2 | weakest notion for G2, strongest for exceptions | via obligation O59 |
| 2004 | Thm 1, 2 | 342, 344 | Level($2^+$) G2; $\Pi_2^*/\Sigma_2^*$ lemmas | full (deferred step); sketch |
| 2005 | §1 | — | Kleene, Rogers, Jeroslow | verbatim (codified §2.2) |
| 2005 | §2–3 | 4–10 | U-Grounding language; Level notions; tableaux; Tab lists; $\mathrm{IS}_D(A)$; Eqs. 5–8 | text layer |
| 2005 | Def 4, 5; Thm 1 | 19–20 | Normed, $\theta$-compactification; engine; $j_2$ proves "the $\Sigma_1^*$ theorem that is $\Phi$'s negation" | full; text layer |
| 2005 | Rem 2, 4 | 21 | 2001 proof hard to visualize; multiplication | —; stated-only |
| 2005 | Lem 1, 2; Thm 2 | 23–26 | *Probe*; non-closure; $\theta = 1/4$; Eq. 20 | full |
| 2005 | fn 3; Thms 3–5; Rem 5; fn 5 | 26–29 | leading negation in neither class; $\mathrm{IS}_D(A)$ consistency-preserving; Part A's U-Height demand | full; text layer |
| 2005 | Defs 2, 3, 6; Lem 3, 4; Thm 6 | 11, 33–36 | $\mathrm{NS}_D^{k,m}$; (27)–(33); Def 6's $q$ proves (25), so step (31) has a gap (ours, §8) | full (Lem 3 sketch); Def 6 image |
| 2005 | Rem 8; p. 37 | 36–37 | stronger Thm 6; $\Pi_1^*$ additions | stated-only |
| 2005 | App A | 39–40 | encoding; (E); uncompressed encodings | text layer |
| 2006a | Thms 1, 2 | 4 | Pudlák's and Solovay's theorems | cited |
| 2006a | Thms 3, 4 | 6–7 | additive naming works; multiplicative fails | full |
| 2006a | Rem 3 | 24 | proof's Gödel number vs. the integer it proves exists | verbatim, image |
| 2006a | App A | 37 | uncompressed encodings "have analogs" | text layer |
| 2007 | Thm 6 | 29 | $T$-1, $T$-2: G2 for tableaux | full |
| 2011 | Rem 6.16 | 28 | "essentially a 1-line proof" | — |
| 2016 | Def 3.4, Ex 3.5 | 7 | self-justification; "ironically typically useless" | verbatim (codified §2.2) |
| 2016 | Rem 4.9; §8 | 13; — | the two questions; "almost cheating" | verbatim (codified §1.4) |
| 2016 | Conj 6.6; Cor 8.2 | 23; 28 | Platonic Stability, conditional | stated-only; text layer |
| 2020 | Xtab; Thm 4.5 | 10; 12 | excluded middle; $\mathrm{IS}_{\mathrm{Xtab}}$ inconsistent | sketch |

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